12

Open industrial Fieldbus and DeviceNet systems

Fieldbus and DeviceNet are communications standards that enable communications between smart or intelligent instruments and a master device such as a PLC. This chapter examines the different Fieldbus systems on the market.

Objectives

When you have completed studying this chapter you will be able to:

- Describe the origin and benefits of Fieldbus and DeviceNet systems
- List and describe the three network classes of Fieldbus and DeviceNet systems
- Describe the characteristics of the following standards (including OSI layers)
  - Actuator sensor interface (AS-i)
  - Seriplex
  - CANbus and DeviceNet
  - Interbus-S
  - Profibus
  - Factory information bus (FIP) and WorldFIP
  - Foundation Fieldbus

12.1 Introduction

There are currently several hundreds of analog and digital standards available for communication between data acquisition and control devices. These field devices communicate using both open and proprietary standards. Traditionally suppliers have produced and sold complete systems that included hardware, software and proprietary protocols. These closed systems made it difficult if not impossible to connect devices of different manufactures. The introduction of open, non-proprietary protocol standards has seen the beginnings of truly open and interoperable systems.

For simplicity the word Fieldbus will be used to refer to both Fieldbus and DeviceNet systems in this chapter. DeviceNet generally refers to the on/off and simpler digital devices whilst Fieldbus tends to encompass instrumentation systems, which need to transfer 16-bit data as a minimum.

A universal open protocol standard is thought by some to be the most desirable conclusion to the problem of multiple Fieldbus systems. The benefits to end users being that all devices would talk using the same protocol and therefore the user could buy any product and plug it in to any system without interfacing problems.

This chapter includes:

- A history of Fieldbus systems
- Classes of Fieldbuses
- The OSI model and Fieldbus systems
- Interoperability
- Examples of various Fieldbus protocols

Before examining the different protocols, it would be helpful to ask why there is considerable effort, time and money being invested in searching for a 'perfect' digital communication network. Why are there several approaches and not just one unified effort? Aren't there enough standards and what is wrong with the one's we have? To answer these questions we need to look at the evolution of digital technology and in particular digital communication technology.

12.2 Overview

Looking at these technologies from a historical perspective, it becomes clear that they are relatively new and, more importantly, still evolving. As technology progresses, more complicated and smaller systems are developed. These new applications and systems reveal shortcomings in the existing technology. This requires the technology to be modified or improved to meet these new demands.

The current approach to cabling a typical control system is shown in Figure 12.1. The concept of Fieldbus is illustrated in Figure 12.2. The figure shows how the instruments are connected with a communication cable. There are numerous benefits not only with regard to minimization of the cables but in greatly increased levels of data available to the operator of the instrument.

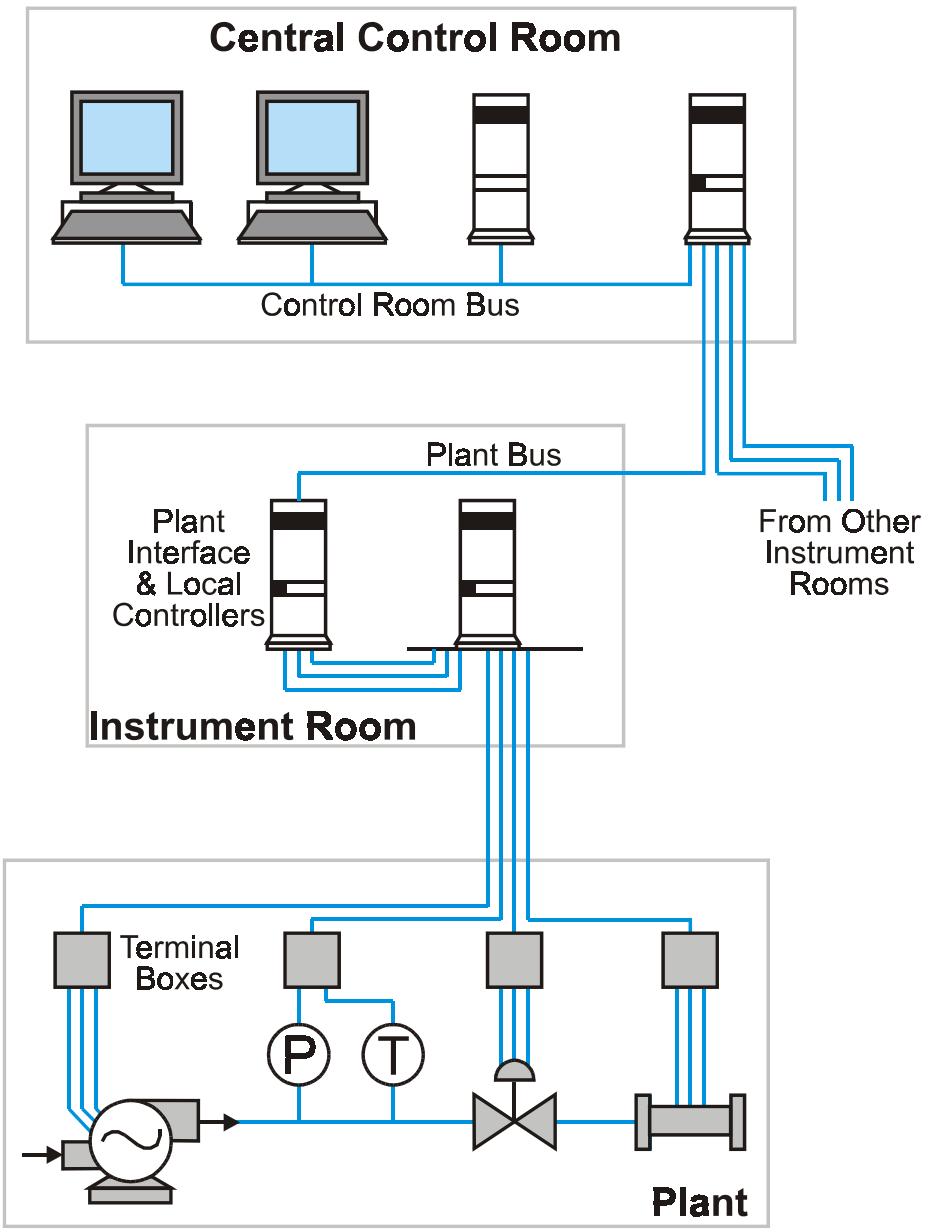


Figure 12.1
Current approach to cabling of a typical control system

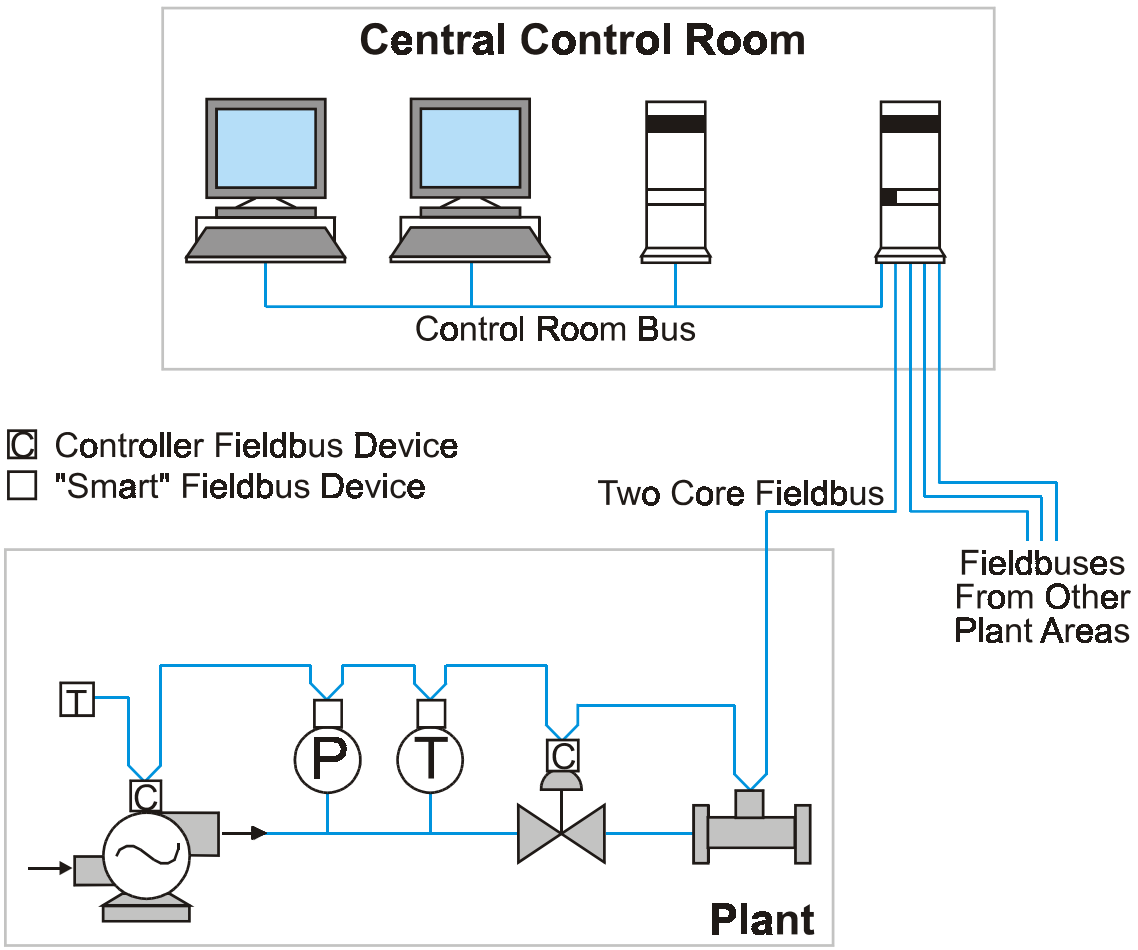


Figure 12.2
Fieldbus approach to cabling of a typical control system

Benefits of the modern Fieldbus

There are real benefits to be gained from the emerging networks, including:

- Greatly reduced wiring costs
- Reduced installation and startup time
- Improved on-line monitoring and diagnostics
- Easy change-out and expansion of devices
- Improved local intelligence in the devices
- Improved interoperability between manufacturers

Classes of Fieldbus networks

It would seem at first that a single Fieldbus system would be beneficial to all users, but this is not the case. Very simple field devices such as proximity switches, limit switches, and basic actuators only require a few bits of digital information to communicate an ‘off’ or ‘on’ state. These are usually associated with real time control applications where update times of a few milliseconds are required. The associated electronics necessary to communicate with these systems can be simple, compact and inexpensive enough to be integrated in the device itself.

Alternatively, complicated devices like PLCs, DCSs, or operator stations (human-machine-interfaces, HMIs), require multi-byte length messages (up to 256 in some systems) and may only require update times of 10–100 ms depending on the application. These systems require larger packets due to a large amount of data to be transferred.

The solution is to select the digital communication network that is best suited to the application, and integrate information up through the higher speed networks as required. Several approaches in digital networks have been developed over the last few years, each with a different target application, speed and technology.

These different approaches are generically referred to as Fieldbus and DeviceNet systems and are typically categorized by the length of the 'message' required by the devices to adequately convey information to the host or network.

This method of categorization allows these Fieldbus and DeviceNet systems to be placed in one of the following three network oriented classes:

- Bit: Sensor level devices such as AS-i
- Byte: Device level instruments such as Interbus-S, CANbus and DeviceNet
- Message: Field level devices such as Profibus and Foundation Fieldbus

Bit oriented systems are used, for example, with simple binary type devices such as proximity sensors, contact closures (pressure switches, float switches, etc.), simple push-button stations and pneumatic actuators. These types of networks are also known as 'sensor bus' networks due to the nature of the devices (sensors and actuators) typically used.

Byte oriented systems are used in much broader applications such as motor starters, bar code readers, temperature and pressure transmitters, chromatographs and variable speed drives due to their larger addressing capability and the larger information content of the several byte length message format.

Message oriented systems, which are those systems containing over 16 bytes per message, are used in interconnecting more intelligent systems such as PCs, PLCs, operator terminals and engineering workstations where uploading and downloading system or device configurations is required, or in linking the above mentioned networks together.

The OSI model and Fieldbus systems

The ISO/OSI is an internationally accepted communications reference model and as such has been universally accepted by all Fieldbus systems committees as a starting point in the design process.

As outlined, the OSI model allocates specific tasks and defines the interface for each layer. The model used in an industrial system is a simplified version with only three layers: application, data link and physical (see Figure 12.3). In addition to the three OSI model levels, a user layer is required in Fieldbus systems, to incorporate the function blocks. This is discussed later.

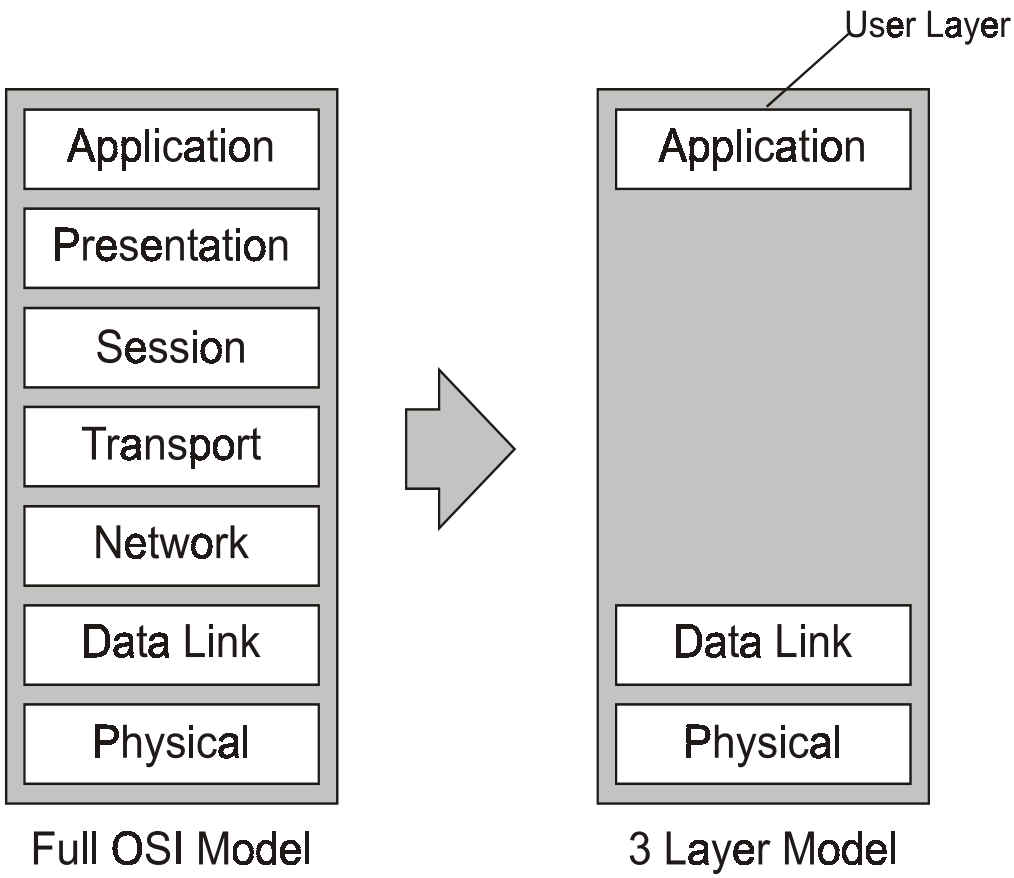


Figure 12.3
OSI and simplified OSI models

The functions of each layer are:

Physical layer

This layer defines the voltages and physical connections. Data received from the data link layer is encoded as electrical information on the actual wire. Similarly, electrical signals received from the wire are passed as binary data to the data link layer.

Data link layer

This defines the protocol and error detection part of the protocol, where the messages sent on the wire are encoded and messages received from the wire are decoded.

Application layer

This layer defines the content messages and the services required supporting them.

Network and transports layers have been omitted by almost every producer of Fieldbus protocols. This means that without a Network layer the protocol cannot ‘internetwork’ as can be done with the TCP/IP protocol. Therefore most industrial Fieldbus protocols are not directly able to communicate over multiple interconnected networks as with Ethernet and TCP/IP.

Interoperability

Interoperability is defined as the capability of using similar field devices from different manufacturers as replacements without losing functionality or sacrificing the degree of integration with the host system. The user is able to choose the right devices for an application independent of the supplier, control system and the protocol.

Refer to Figure 12.4.

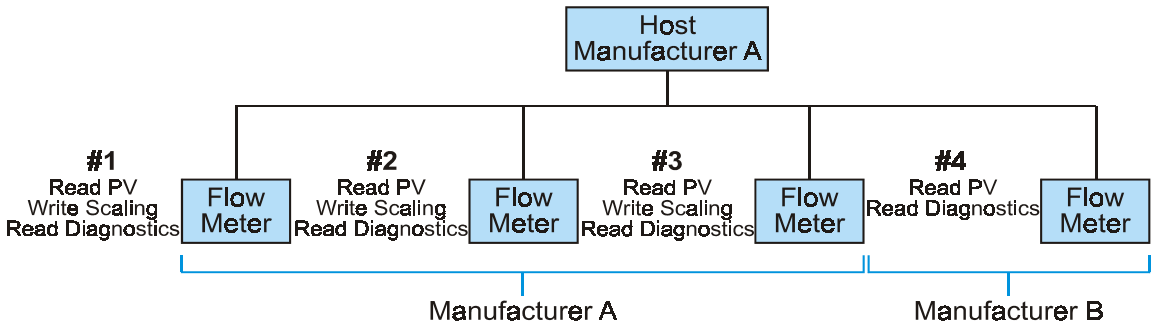


Figure 12.4
A Non-interoperable system

The host system, from manufacturer A, can access flow meters at addresses 1, 2 and 3 from manufacturer A with full read/write capability, but only has read capability for the flow meter at address 4, from manufacturer B. Therefore, the host control system treats each of these field devices differently and they could not be used as effective replacements for each other. Only if the flow meter at address 4 is totally interchangeable with the other devices is the system considered interoperable.

Interoperability is valuable because:

- It allows the end user to select different manufacturers' devices in an interchangeable manner with no discernible differences in the use of each device.
- The concept allows the easy integration of new field devices into control strategies, as they become available.

Importantly, a communication hierarchy, such as the OSI model, cannot address the issue of interoperability. Standardization of the physical, data link and application layers will ensure information can be exchanged among devices on a Fieldbus network. It is the User layer that actually specifies the type of data or information and how it is to be used. Hence, specification of the user layer is vital to ensure complete performance of a Fieldbus system (although it is not part of the OSI communications model).

Protocol review

The following sections include a short review of selected open Fieldbus standards. These include:

- Actuator sensor interface (AS-i)
- Seriplex
- CANbus, DeviceNet and SDS
- Interbus-S
- Profibus
- Foundation Fieldbus

12.3 Actuator sensor interface (AS-i)

The AS-i is a master/slave, open system network developed by eleven manufacturers. These manufacturers created the AS-i Association to develop an open Fieldbus specification. Some of the more widely known members of the AS-i association include Pepperl-Fuchs, Allen-Bradley, Banner Engineering, Datalogic Products, Siemens, Telemecanique, Turck, Omron, Eaton, and Festo. The number of AS-i Association members continues to grow. The AS-i Association also certifies that products under development for the network meet the AS-i specifications. This will assure compatibility between products from different vendors.

AS-i is a bit oriented communication link, designed to connect binary sensors and actuators. Most of these devices do not require multiple bytes to adequately convey the necessary information about the device status, so the AS-i communication interface is designed for bit-oriented messages to increase message efficiency for these types of devices.

The AS-i interface is an interface for binary sensors and actuators, designed to interface binary sensors and actuators to microprocessor based controllers using bit length 'messages'. It was not developed to connect intelligent controllers together as this would be far beyond the limited capability of short bit length message streams.

Modular components form the central design of AS-i. Connection to the network is made with unique connecting modules requiring minimal, or in some cases no tools, and provide for rapid, positive device attachment to the AS-i flat cable. Provision is made in the communications system to make 'live' connections, permitting the removal or addition of nodes with minimum network interruption.

Connection to higher-level networks is made possible through plug-in PC, PLC cards or serial interface converter modules.

The following sections examine these features of the AS-i network in more detail.

The physical layer

AS-i uses a two-wire untwisted, unshielded cable, which serves as both communication link and power supply for up to thirty-one slaves. A single master module controls communication over the AS-i network, which can be connected in various configurations such as bus, ring, or tree (see Figure 12.5). The AS-i flat cable has a unique cross-section that permits only properly polarized connections when making field connections to the modules (see Figure 12.6). Other types of cable may be used for the AS-i network providing they meet the AS-i cable specification. A special shielded cable is also available for high noise environments.

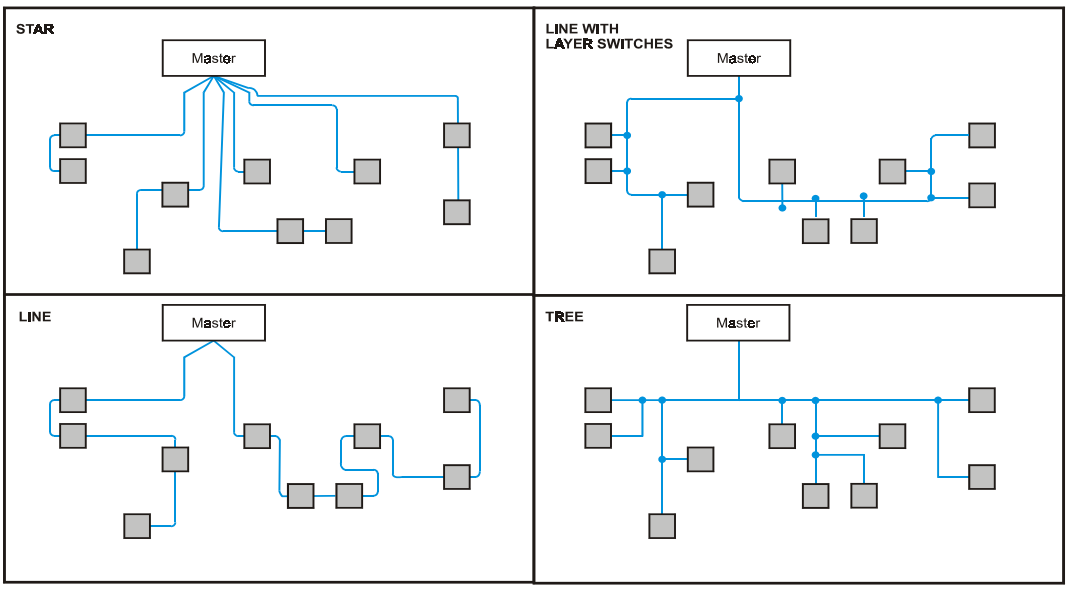


Figure 12.5
AS-i topographical examples

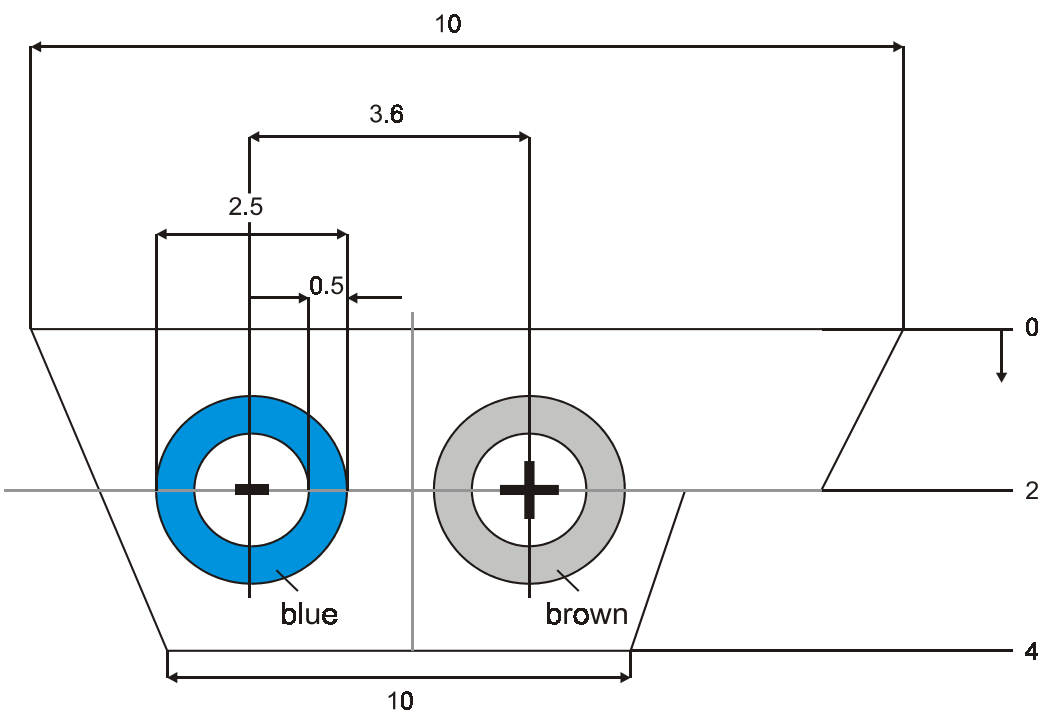


Figure 12.6
AS-i cable cross-section

Each slave is permitted to draw a maximum of 65 mA from the 30 V dc power supply. If devices require more than this, separate supplies must be provided for each such device. With a total 31 slaves drawing 65 mA, a total limit of 2 A has been established to

prevent excessive voltage drop over the 100 m permitted network length. A 16 AWG cable is specified to ensure this condition.

The slave (or field) modules are available in four configurations:

- Input modules for 2- and 3-wire dc sensors or contact closure
- Output modules for actuators
- Input and output (I/O) modules for dual purpose applications
- Field connection modules for direct connection to AS-i compatible devices

The I/O modules are capable of accepting up to 4 I/O per slave, and a total of 124 I/O for the network.

A unique design allows the field modules to be connected directly into the bus while maintaining network integrity (see Figure 12.7). The field module is composed of an upper and lower section; secured together once the cable is inserted. Specially designed contact points pierce the self-sealing cable, providing bus access to the I/O points and/or continuation of the network. True to the modular design concept, two types of lower sections and three types of upper sections are available to permit 'mix-and-match' combinations to accommodate various connection schemes and device types (see Figure 12.8). Plug connectors are utilized to interface the I/O devices to the slave (or with the correct choice of modular section screw terminals) and the entire module is sealed from the environment with special seals where the cable enters the module. The seals conveniently store away within the module when not in use.

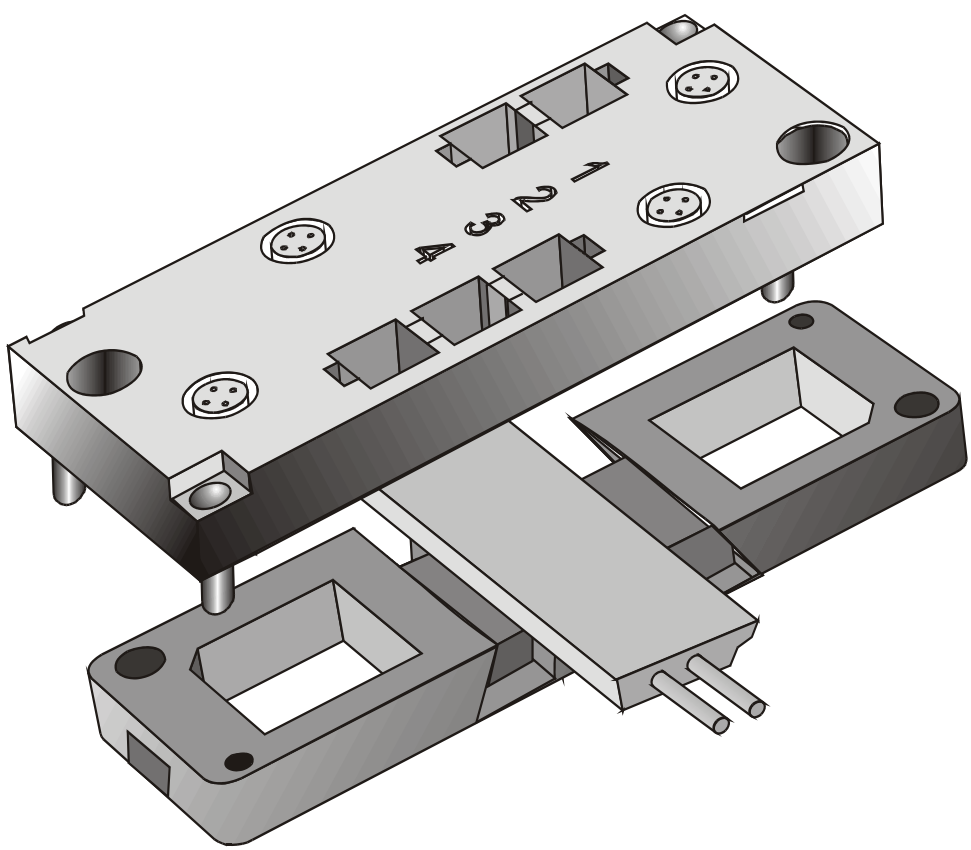


Figure 12.7
AS-i cable to device connections (1)

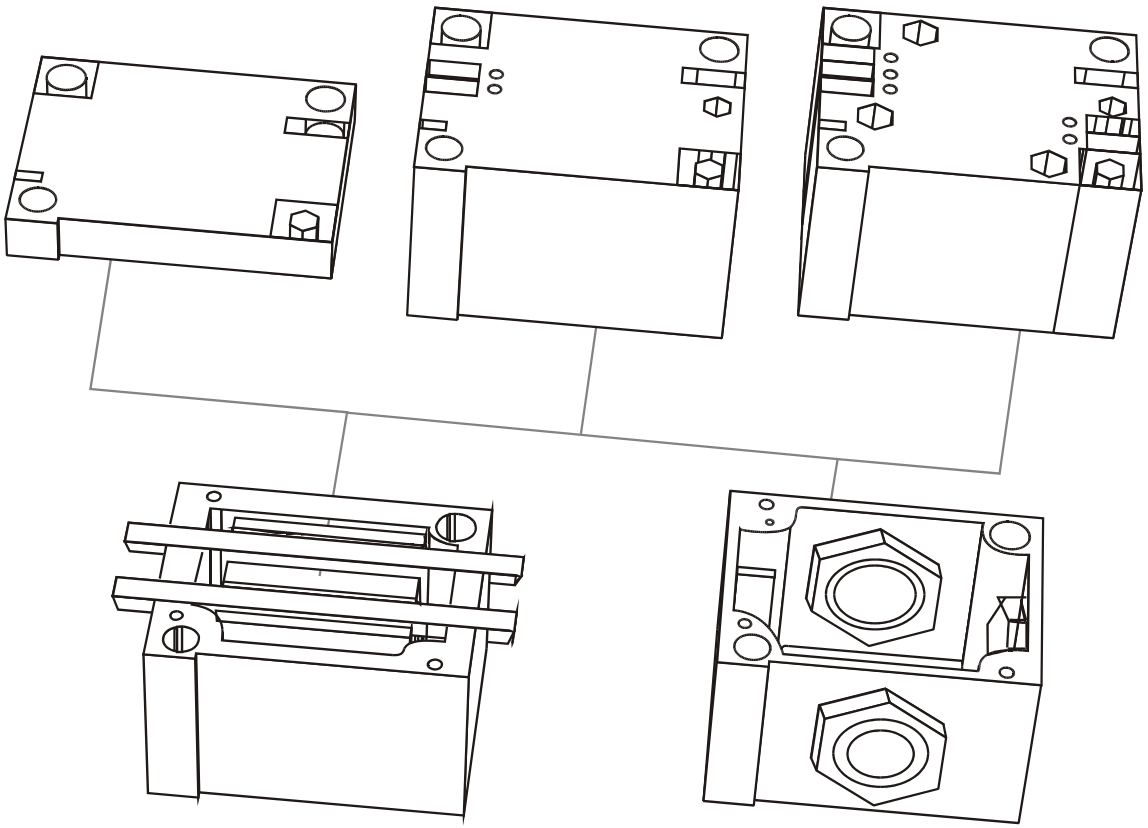


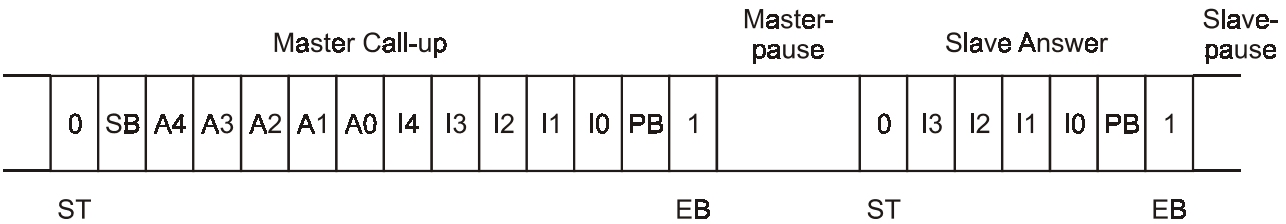
Figure 12.8
AS-i cable to device connections (2)

The AS-i network is capable of a transfer rate of 167 kbps. Using an access procedure known as ‘master-slave access with cyclic polling’, the master continually polls all the slave devices during a given cycle to ensure rapid update times. For example, with all 31 slaves and 124 I/O points connected, the AS-i network can ensure a 5 ms cycle time, making the AS-i network one of the fastest available.

A modulation technique called ‘alternating pulse modulation’ provides this high transfer rate capability as well as high data integrity. This technique is described in the following section.

The data link layer

The data link layer of the AS-i network consists of a master call-up and slave response. The master call-up is exactly fourteen bits in length while the slave response is 7 bits. A pause between each transmission is used for synchronization, error detection and correction. Refer to Figure 12.9 for example call-up and answer frames.



ST	Startbit:	marks the beginning of a master call-up =0: valid start bit =1: not allowed
SB	Control Bit:	marks the data/parameter/address call-ups respectively the command call-up =0: data/parameter/address call-up =1: command call-up
A0...A4	Address:	address of the called slave (5 bit)
I0....I4	Information	the 5 information bits contain the information per each call-up type, which is communicated to the ASI slave. Details are described for each current report.

Figure 12.9
AS-i packet format

Various code combinations are possible in the information portion of the call-up frame. It is these code combinations that are used to read and write information to the slave devices. Examples of some of the master call-ups are listed in Figure 12.10. A detailed explanation of these call-ups is available from the AS-i association literature and is only included here to illustrate the basic means of information transfer on the AS-i network.

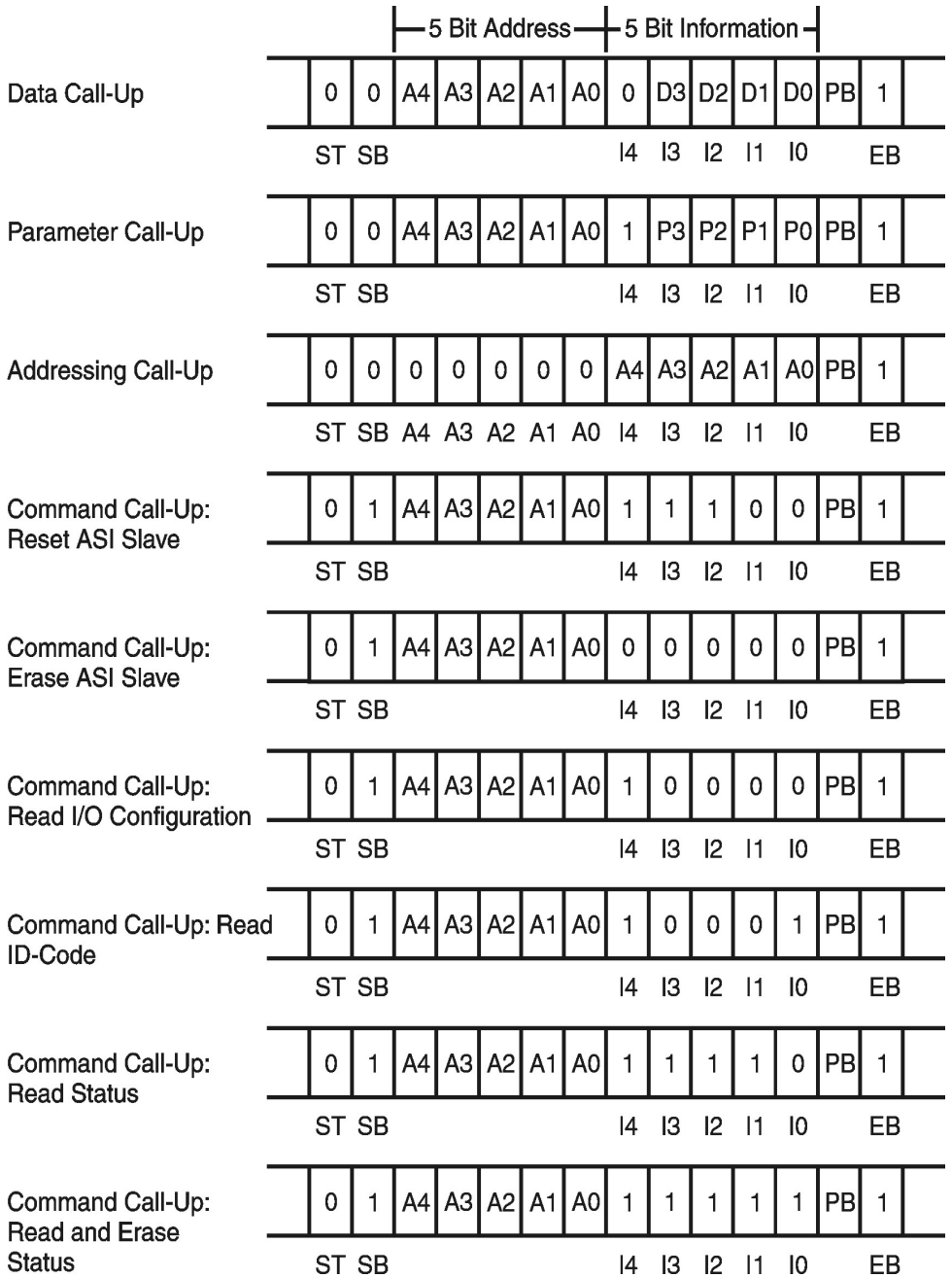


Figure 12.10
AS-i packet format continued

The modulation technique used by AS-i is known as ‘alternating pulse modulation’ (APM). As the information frame is of a limited size, providing conventional error checking was not possible and therefore the AS-i developers chose a different technique to insure high-level data integrity.

Referring to Figure 12.11, the coding of the information is similar to Manchester II coding, but utilizing a ‘sin squared’ waveform for each pulse. This wave-shape has several unique electrical properties, which reduce the bandwidth required of the

transmission medium (permitting faster transfer rates), and reduce the end of line reflections, common in networks using square wave pulse techniques. Also, each bit has an associated pulse during the second half of the bit period. This property is used as a bit level of error checking by all AS-i devices. The similarity to Manchester II coding is due to this technique having been used for many years to pass synchronizing information to a receiver along with the actual data.

In addition, AS-i developers also established an internal set of regulations for the APM coded signal, which is used to further enhance data integrity. For example, the start bit or first bit in the AS-i telegram must be a negative impulse and the stop bit a positive impulse. Two subsequent impulses must be of opposite polarity and the pause length between two consecutive impulses should be 3 ms. Even parity and a prescribed frame length are also incorporated at the frame level. So the 'odd' looking wave form, combined with the rules of the frame formatting, the set of regulations of the APM coded signal, and parity checking, work together to provide timing information and a high level of data integrity for the AS-i network.

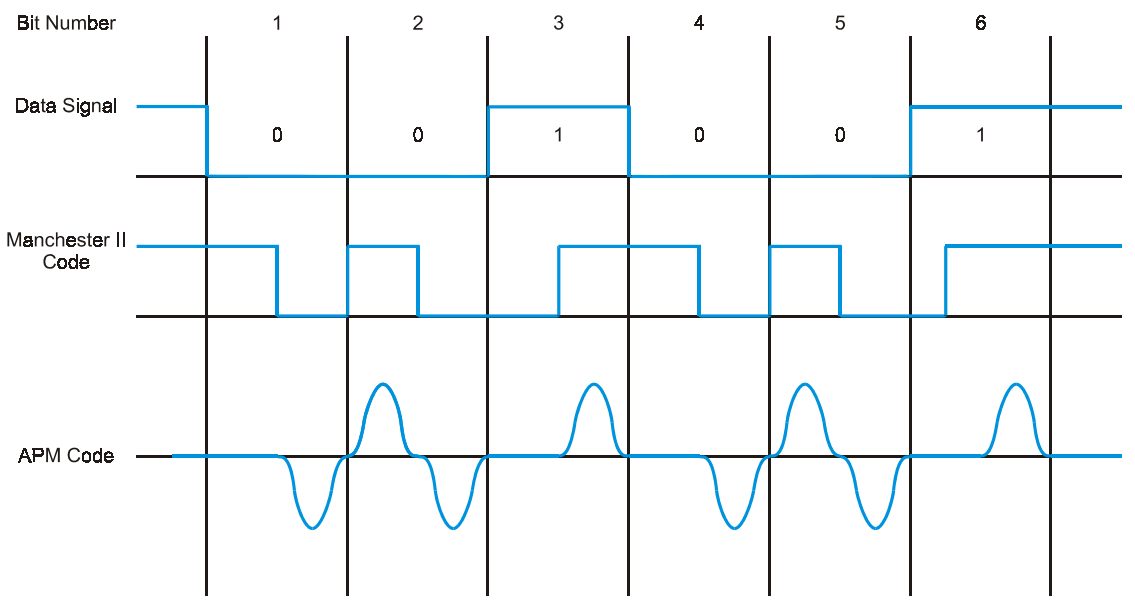


Figure 12.11
Alternating pulse modulation

Operating characteristics

AS-i node addresses are stored in non-volatile memory and can be assigned either by the master or one of the addressing or service units. Should a node fail, AS-i has the ability to automatically reassign the replaced node's address and in some cases reprogram the node itself allowing rapid response and repair times.

Since AS-i was designed as an interface between lower level devices, connection to higher-level systems enables the capability to transfer data and diagnostic information. Plug-in PC cards and PLC cards are currently available. The PLC cards allow direct connection with various Siemens PLCs. Serial communication converters are also available to enable AS-i connection to conventional RS-232, 422, and 485 communication links. Direct connection to a Profibus field network is also possible with

the Profibus coupler, enabling several AS-i networks access to a high-level digital network.

Handheld and PC-based configuration tools are available which allow initial start-up programming and also serve as diagnostic tools after the network is commissioned. With these devices on-line monitoring is possible to aid in determining the health of the network and locating possible error sources.

12.4 Seriplex

Automated Process Control, Inc. developed the Seriplex control bus in 1987 specifically for industrial control applications. The Seriplex Technology Organization Inc. was formed to provide information concerning Seriplex, distribute development tools and the Seriplex Application Specific Integrated Circuit (AS-iC) chip, as well as technical assistance for Seriplex developers. Like other sensor (bit level) networks Seriplex was designed to interface lower level I/O devices over a dedicated cabling system, while providing the capability to connect to a host controller or higher-level digital networks. However, for simple control functions, a unique feature of the Seriplex network allows configuration in a peer-to-peer mode that does not require a host or supervisory controller.

Seriplex allows the implementation of simple control schemes without the need for a supervisory processor. This is done through the use of intelligent modules providing a link between inputs and outputs similar to logic gates, i.e. outputs can be programmed based on the status of certain inputs. If more complicated control is required, or supervisory functions desired, Seriplex may be connected to a host processor through interface adapters. Various PLC and PC plug-in cards are available for this interface (see Figure 12.12).

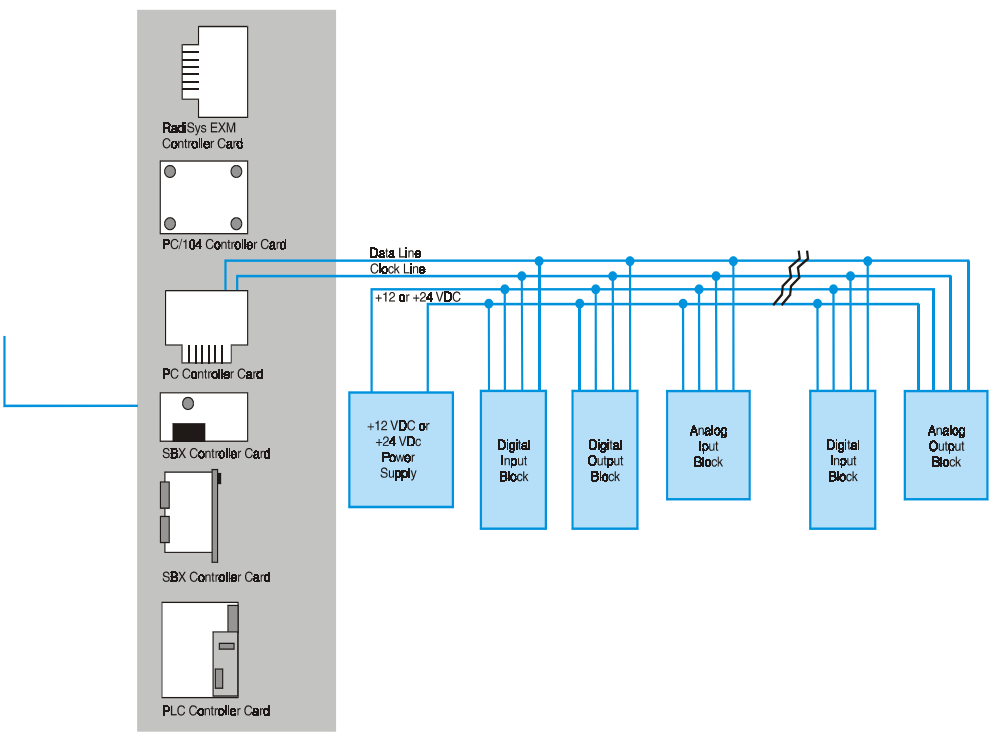


Figure 12.12
Seriplex system example

Various physical topologies are possible for connecting the modular components of the Seriplex network via a five-conductor cable, which provides power, data communications – and clocking signals. Over 7,000 binary I/O points or 480 analog channels (240 In, 240 Out) or various combinations can be supported by Seriplex over this cabling system. The basic configuration without multiplexing can support 255 digital I/O, 32 analog I/O or some combination thereof.

The following sections describe the Seriplex network in more detail.

The physical layer

The Seriplex cabling system consists of a single four-conductor cable with two AWG #22 shielded wires for data and clock signals and two AWG #16 wires for power and common. A shield drain wire is also provided for shield grounding. Clock rates from 16 to 100 kHz are selectable with newer versions capable of up to 200 kHz. Capacitance values of cable dramatically affect all communication systems and low capacitance cable designs are available from several manufacturers to maximize data transfer rates. Rates of up to 100 kHz over 500 feet are possible with Seriplex using low capacitance (16 pF/ft) cabling. However, 20 pF/ft cables would limit this distance at 100 kHz to 350 ft. The importance of low capacitance cabling cannot be over emphasized in any system.

12 V dc is provided by the cable to power the I/O devices in the first generation systems. Second generation systems operate on either 12 or 24 V dc, with the level selected by the user for the particular system used. Field connections are made through Seriplex modules located near the field devices.

Individual I/O addresses are programmed in the module to allow the network access to each point. A total of 255 usable addresses is available to the modules. Digital inputs and outputs use one address each. Each 8-bit analog module uses eight addresses (for one analog input or output). Multiplexing methods are employed to increase the total digital I/O count to 7,706 or analog I/O count to 480 or a combination of these.

Data and clock signals are transferred over the network in the form of 0 to +12 V digital pulses.

The data link layer

Two different methods of operation are possible with Seriplex, depending on the mode of operation. Both modes of operation use the unique access control method described in Mode 2 below.

In Mode 1, or peer-to-peer mode, modules can be logically inter-linked without the need for a host controller. In this case logical functions are implemented directly between modules. A separate clock module is required in this mode as there is no host to provide crucial clock line information. Module outputs can be logically programmed to function, based on the status of other modules' inputs. With this capability simple logic functions can be performed without the need of a host controller.

Mode 2 operation requires the host controller to provide timing clock signals. The receiver in each module counts the clock pulses. When the pulse count of the clock line equals the receiver's address, access is granted on the data link line for the receiver to read from and, in turn, write to, the host controller.

This access control method is unique in that a continuous 'train' of clock and data pulses cycle through the system. Access on the data line for individual addresses (bit status) is granted for a time period within the data stream based on the time slot of the address (see Figure 12.13). This 'continuous polling' starts with a synchronizing signal 8 clock cycles long and serves as notification that the 'polling' is about to begin. At the

beginning of the cycle the data line is 'empty'. As the pulse count equals each module's address count, the modules 'dump' their bit values on the data line so that at the end of the cycle all information is available to the host. The frame size can be adjusted in length from 16 to 256 bits, in multiples of 16, to accommodate different size systems. Correct sizing of the system and resultant frame size can provide extremely fast update times for smaller networks.

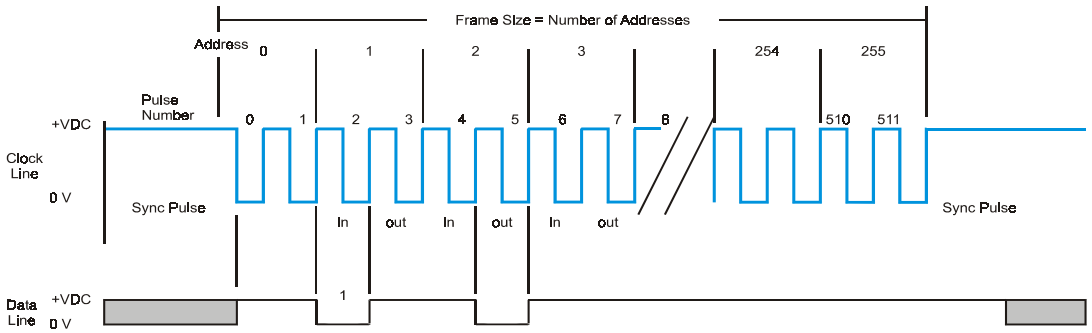


Figure 12.13a
Seriplex mode 1

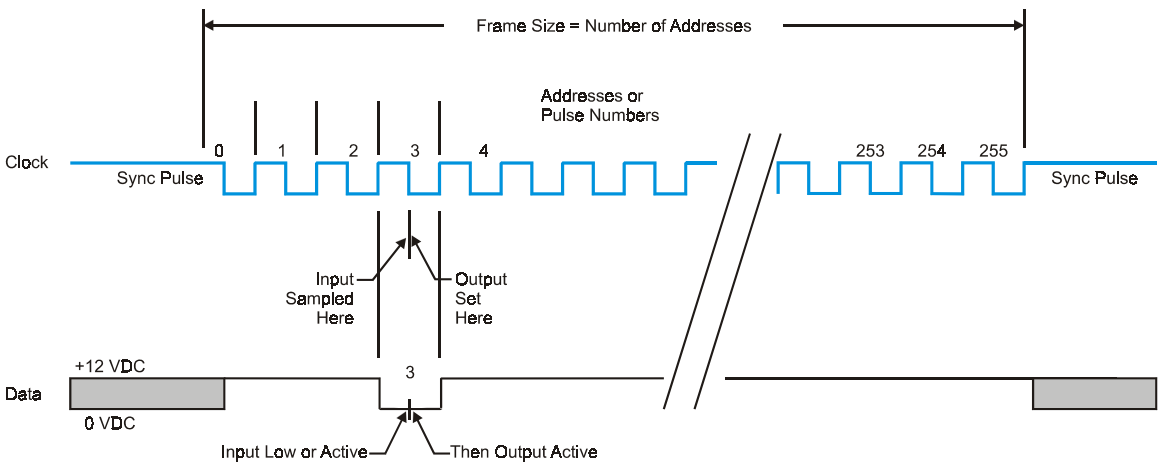


Figure 12.13b
Seriplex mode 2

Data echoing provides error detection on the bit level. The receiver echoes messages (which are typically one bit per address) to confirm correct data receipt. This is not automatic and is implemented in the application layer of the software by the applications programmer.

Operating characteristics

Seriplex is a bit oriented network system intended to link lower level devices both physically and, in Mode 1 operation, logically. These features are incorporated in the AS-I-C chip located in all Seriplex devices. Handheld programming devices are available to enable Seriplex device configuration.

Interface devices to higher-level field networks through host controllers or special Gateways are also available.

12.5 CANbus, DeviceNet and SDS systems

CANbus

The CAN network was developed in the automotive industry in response to the rapidly growing use of electronic-control systems in automobiles. As demands for fuel efficiency and safety increased, more and more electronic devices became part of the system. The need for multiple devices to pass information between them rapidly became a necessity. A type of serial data bus system was developed by Bosch to meet these demands. It was called the controller area network or CAN. CAN is formally specified in: 1) BOSCH CAN specification – Version 2.0, Part A, and 2) ISO 11898: 1993 – road vehicles – Interchange of digital information – controller area network (CAN) for high-speed communication. CAN has since been rapidly adapted to industrial applications.

CAN is a bus type network system which does not use a bus master or token passing schemes to access the bus. Instead it uses a unique access control method called 'non-destructive bit-wise arbitration'. This type of access control uses the station identifier bit pattern itself to gain access to the bus as shown in Figure 12.14. The priority of the station is determined by the addressing assignments during configuration of the network and allows the station with the highest priority preferred access. Unlike token passing or master-slave type arbitration schemes CAN is not deterministic, but defers to the station with highest priority making the lower priority stations wait for access.

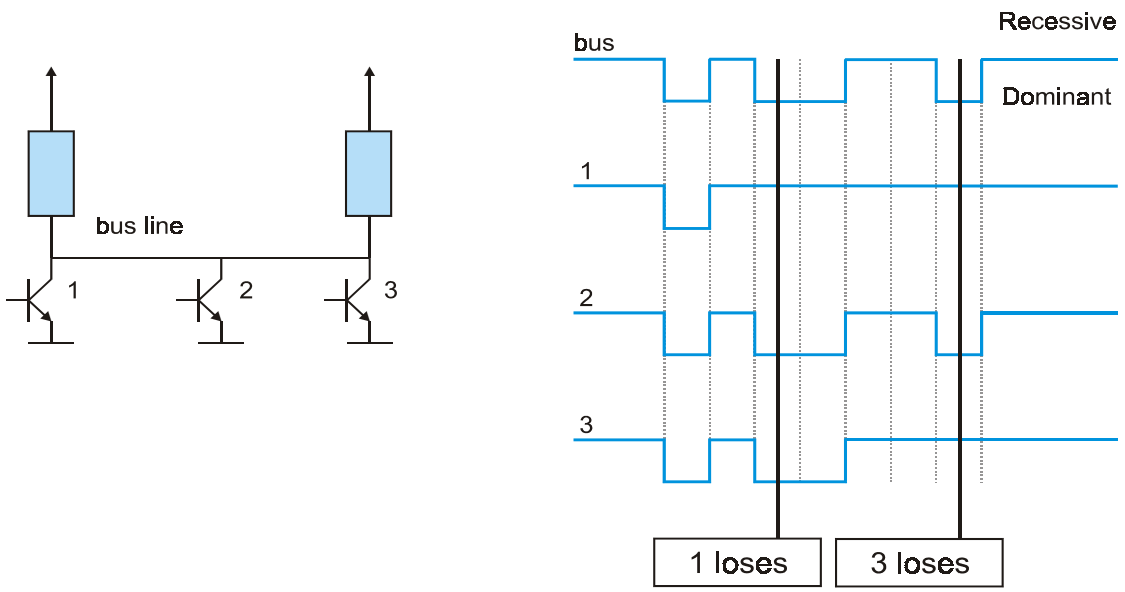


Figure 12.14
Bit arbitration example

Figure 12.14 shows the bit arbitration of a CAN type system. The devices 1, 2, and 3 try to transmit at the same time. Ground or a '0' is dominant. The results can be seen on the top waveform. Because device 1 puts out a '1' and it is dominated by the '0' from 2 and 3, it loses and stops transmitting. Then device 3 puts out a '1' and is dominated by 2. Therefore, 2 continues to transmit while 1 and 3 wait until the line is clear.

The CAN station that wins the arbitration continues to transmit its message frame uninterrupted with no corruption from the other stations arbitration attempts. This allows higher efficiency of data transfer over the network. A typical CAN message frame is shown in Figure 12.15. Notice the data field can be of variable length – up to 8 bytes. This makes CAN suitable for use with more sophisticated devices that may require several bytes to adequately convey their information content. CRC error checking and specific frame length requirements, as well as individual message acknowledgments, are used to ensure data integrity over the bus.

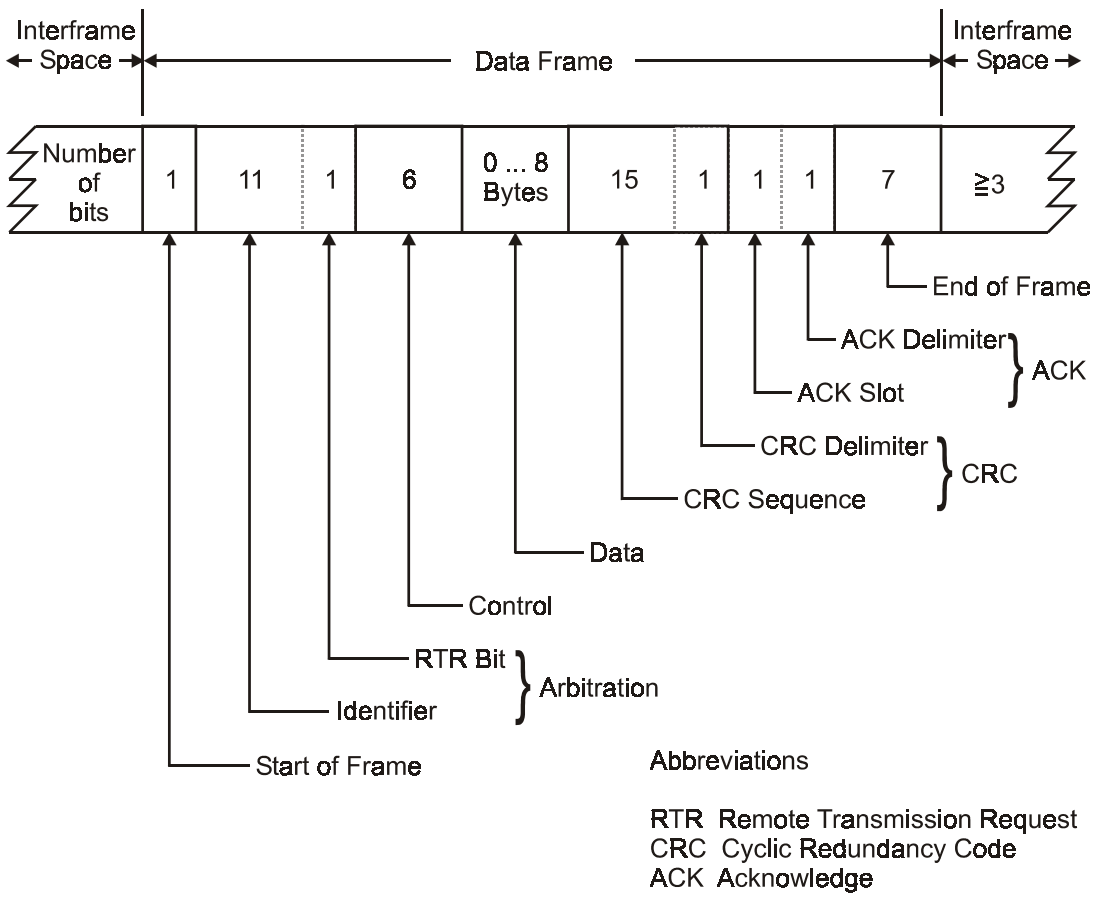


Figure 12.15
The canbus packet

The CAN protocol specifications cover only the physical (layer 1) and data link (layer 2) layers of the ISO/OSI model. Specifics concerning the physical medium for the communication link and the application layer (layer 7) are left to the designers of the systems as described below.

DeviceNet

DeviceNet, developed by Allen-Bradley, is a low-level device oriented network based on the CAN network. It is designed to interconnect lower level devices (sensors and actuators) with higher-level devices (controllers). The variable, multi-byte format of the

CAN message frame is well suited to this task as more information can be communicated per message than with the bit type systems.

The Open DeviceNet Vendor Association Inc. (ODVA) has been formed to issue DeviceNet specifications, ensure compliance with the specifications and offer technical assistance for manufacturers wishing to implement DeviceNet. Over 125 firms have either joined formally or have signed intent to become members. The DeviceNet specification is an open specification and available through the ODVA.

DeviceNet can support up to 64 nodes, supporting as many as 2048 total devices. A single, four-conductor cable provides both power and data communications. Various devices are available to interconnect I/O devices, and the network trunk-line cable allowing customized configurations.

As DeviceNet was designed to interface lower level devices with higher-level controllers, a unique adaptation of the basic CAN protocol was developed. This is similar to the familiar poll/response or master/slave technique, but still utilizes the speed benefits of the original CAN.

Figure 12.16 shows the DeviceNet profile in its relationship to the ISO/OSI Model. It is important to note that only Layers 1 and 2 are covered by the CAN protocol specification while the remaining layers were developed for the DeviceNet network.

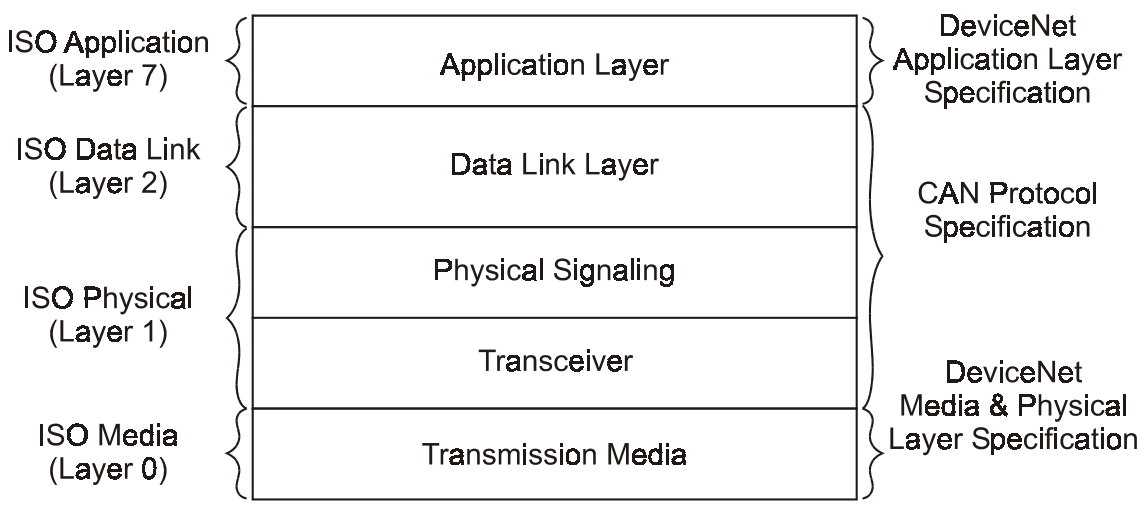


Figure 12.16
DeviceNet and the OSI model

The following sections examine these features of the DeviceNet network and protocol in more detail.

The physical layer

The DeviceNet cabling system consists of a single four-conductor cable in a bus topology providing both power and data communication. Data is transmitted over the #18 twisted pair. Power is provided over the #15 twisted pair. Both pairs have a foil shield and an overall braid with #18 drain wire. Terminating resistors are required on both ends of the trunk line. 24 V dc power is provided on the power bus and can support up to 3 Amp on the DeviceNet thin cable or 8 amp on the DeviceNet thick cable. The total length of trunk line allowed depends on which type of cable is used, the number of devices supported and the data rate. The voltage at each device should be at least 11 V dc or higher.

Data rates of 125, 250, and 500 kbps are possible with the corresponding network configuration shown in Figure 12.17. Various connectors can be used to connect devices to the network such as screw terminals, screw terminals or sealed screw tight connectors.

Data Rate	Trunk Distance	Drop Length	
		Maximum	Cumulative
125 k baud	500 meters (1600 ft.)	3 meters (10 ft.)	156 meters (512 ft.)
250 k baud	200 meters (600 ft.)		78 meters (256 ft.)
500 k baud	100 meters (300 ft.)		39 meters (128 ft.)

Figure 12.17
DeviceNet lengths and baud rates

To allow non-destructive arbitration during simultaneous transmission from two or more nodes the BOSCH CAN specification defines the two possible logic levels as ‘dominant’ and ‘recessive’. During arbitration the dominant value will win access to the bus. For DeviceNet, the dominant level is represented by a logical ‘0’ and the recessive level by a logical ‘1’. The electrical voltage levels representing these logic levels are taken from the ISO 11898 standard. CAN utilizes a balanced transmission system with data signals appearing as the difference between CAN_H and CAN_L.

DeviceNet specifications require isolation to prevent ground loops. As the circuitries in all devices are ultimately referenced to the V-bus signal, connection of the network should be earth grounded at the bus power supply only. All devices attached to the network must either be referenced to V– or otherwise ground isolated.

DeviceNet requires the following features to be incorporated within the physical and media layers:

- Use of CAN technology
- Support of both thick and thin drop line
- Ability to operate at a minimum of three data rates
- 125 kbaud for distances up to 500 meters (max.) (1640 feet)
- 250 kbaud for distances up to 200 meters (max.) (656 feet)
- 500 kbaud for distances up to 100 meters (max.) (328 feet)
- Linear bus topology
- Low loss and low delay cable
- Shielded twisted pair cable, containing both power and signal pairs
- Small size and low cost
- Support of up to 64 nodes
- Support drop lines of up to 6 meters (20 feet) in length
- Node removal without disturbing or interrupting network operation
- Simultaneous support of isolated and non-isolated physical layers

DeviceNet uses two types of pre-made cables, thick cable and thin cable. The thick cable is a large gray cable used as long trunk runs between devices. The thin cable is a small and usually short yellow cable that connects thick cables to devices. The thick and thin cables are connected together by a large black ‘T’ junction. All cables and connections have threaded ring connectors.

The data link layer

The data link layer is specified in the CAN protocol specification (see Figure 12.16). The format of the data link layer (frame format) is fixed by this specification. However, the method used to encode the identifier and data fields in the CAN message packet is left to the application layer developer as described in the following section. The method of communication is based on the producer/consumer approach where one station (*the producer) places data on the bus at regular intervals and this is then read by the consumer station on the network.

The application layer

The CAN specification does not dictate how information within the CAN message frame fields are to be interpreted – this was left up to the developers of the specific application software. In the case of DeviceNet a unique method was developed to allow for two types of messages to exist.

Through the use of special identifier codes (bit patterns), master is differentiated from slave. Also, sections of this field tell the slaves how to respond to the master's message. For example, slaves can be requested to respond with information simultaneously in which case the CAN bus arbitration scheme assures the timeliest consecutive response from all slaves in decreasing order of priority. Or, slaves can be polled individually, all through the selection of different identifier field codes. This technique allows the system implementers more flexibility when establishing node priorities and device addresses.

System operation

Several devices are available to allow connection of DeviceNet to higher-level devices. For example, Allen-Bradley has developed PLC plug in cards to function as DeviceNet scanners. These devices support a master/slave configuration communicating with slave devices through either the strobe or poll methods. Two separate DeviceNet channels (or networks) can be supported. These modules also perform limited diagnostics on the network, and chassis communication link report this information to higher-level controllers. An interface is also available, which allows a PC to act as another node on the Network.

With the DeviceNet flex I/O adapter up to 128 non-DeviceNet compatible devices can communicate to other DeviceNet I/O and PLC controllers. Other types of DeviceNet compatible products are also being marketed which are connected directly to the network with a minimum of configuration effort.

Smart distributed system (SDS)

The smart distributed system (SDS) was developed by Honeywell and is a low-level device oriented network based on the CAN network. It is designed to interconnect lower level devices (sensors and actuators) with higher-level devices (controllers). The variable, multi-byte format of the CAN message frame is well suited to this task since more intelligence can be communicated per message than with the bit type systems.

The SDS 'partners' program has been formed, and in cooperation with Honeywell issues the SDS specifications, ensures compliance and offers technical assistance for manufacturers wishing to implement SDS. The SDS specification is an open specification and available through Honeywell and the SDS 'partners' program.

The SDS network can connect up 126 devices on a single bus. Each group of 16 I/O is interfaced to a higher-level device (PLC, for example) through the interface terminal strip

(ITS). The ITS provides the physical interface between the network bus and individual I/O points on the PLC I/O cards. Plug-in cards are also available to interface the bus directly to the PC. This choice between interfaces gives the designer a method for integrating the SDS with an existing PLC system.

The SDS utilizes the CAN network to allow devices to report information only when there is a need, e.g. a change of state of an input to the controller. This approach reduces traffic on the network by minimizing polling inquires from the controller to the slave devices.

As with other CAN based systems the SDS network uses the OSI Layers 1 and 2 (physical and data link layers) of the CAN protocol and develops the SDS application layer (OSI Layer 7) for its specific target application area, integrating lower level devices with higher level controllers.

The following sections examine these features of the smart distributed system (SDS) network and protocol in more detail.

The physical layer

The SDS cabling system consists of a single, four conductor, shielded, cable in a bus topology providing both power and data communication. Both data and power pairs are twisted and an overall shield is provided for noise protection. Terminating resistors are required on both ends of the bus. 12 to 24 V dc power is provided on the power bus to support field devices. The total length of trunk line allowed depends on which type of cable is used, the number of devices supported and the data rate.

Various data rates are possible with the corresponding network configuration restrictions. Several connector types can be used to connect devices to the network such as screw terminals or sealed screw tight connectors.

The data link layer

The data link layer is specified in the CAN protocol specification (see Figure 12.16). The format of the data link layer (frame format) is fixed by this specification. However, the method used to encode the identifier and data fields in the CAN message packet is left to the application layer developer.

The application layer

The CAN specification does not dictate how information within the CAN message frame fields is to be interpreted; this was left up to the developers of the specific application software. In the case of SDS, various codes within the identifier frame allow for communication between slave devices and controllers.

Through the use of special identifier codes (bit patterns) unique addresses are established for each device. Source and destination addresses of messages are distinguished by the setting (1 or 0) of the most significant CAN identifier bit, called the SDS Direction bit. A 0 designates the address, what follows is the destination; a 1 designates it as the source of the message. The application protocol allows any device, which needs to read the message (or ‘consumer’) access to this information as it appears on the network. There can be more than one consumer of a given message.

Conversely, when a device senses a change of state it can put that information on the network as soon as it can gain access to the bus consistent with the CAN arbitration procedure. This device is known as a ‘producer’ in the CAN protocol.

Through these unique CAN identifier field code patterns, SDS provides the functions and unique capabilities of this flexible and fast device level networking system.

System operation

Several features of SDS are implemented at the system level to speed startup and monitor the 'health' of the devices and network. One of these is the Autobaud. Through this special function of the bus manager, (a designated device controller on the network, usually the host controller) a unique message packet is sent immediately after initial bus power-up. This allows all the other devices to monitor the time length of the frame and determine the baud rate setting of the controller. Each device can then adjust its baud rate accordingly to ensure all devices operate at the same data rate.

Continuous monitoring for 'missing' devices and defective devices is implemented by periodic polling. If a device fails to report within a specified period of time the host will flag the device missing warning. Polling will continue until the device reappears.

Another monitoring feature is the periodic device self-test control. Periodically, instead of device polling, the host will substitute the self-test command and examine the device's diagnostic registers in the reply for device errors.

Several devices developed through the SDS 'partners' program have enabled direct SDS connection to various higher level devices such as PLCs, PCs, VMEbus systems, starters and pilot devices. Interfaces to new devices are certainly possible in the future for SDS as this network continues to find new applications in the industry.

12.6 Interbus-S

The Interbus-S is an open device level network that allows connection of up to 4096 digital I/O points over a distance of up to 400 m. Through a unique frame transfer protocol these points can be updated in as little as 14 ms, faster speeds are possible with a lower I/O count. It is a timed ring topology with subsystem drops (tree structure) allowing connections of up to 256 stations. Data rate is 500 kbps.

The variable length frame format allows message frames of up to 512 bytes enabling communication between intelligent I/O devices. Integration to the higher-level Fieldbus networks is also within the capability of this network.

The Interbus-S Club, founded in 1993 was established to maintain and advance the Interbus-S network standard. The organization provides Interbus-S specifications to potential developers and assists with technical information.

The following sections examine these features of the Interbus-S (IBS) network and protocol.

The physical layer

The Interbus-S cabling system specification allows for either twisted pair copper or fiber optic cable connected to each station in a ring topology. Communication is serial and frame transmission is accomplished through a unique register shifting procedure developed specifically for the Interbus-S network. Two types of communication buses are used as part of the same network – local bus and remote bus. Each bus type carries the same signals but at different electrical levels. Local bus operates at TTL voltage levels and is designed for short distances typically within a control enclosure. Remote bus utilizes RS-485 voltage levels and is designed to communicate over much longer distances – up to 1300 ft (400 m). Both buses operate at 500 kbps transfer speed and a

special module; the BK module is required to translate the two signal levels (Figure 12.18).

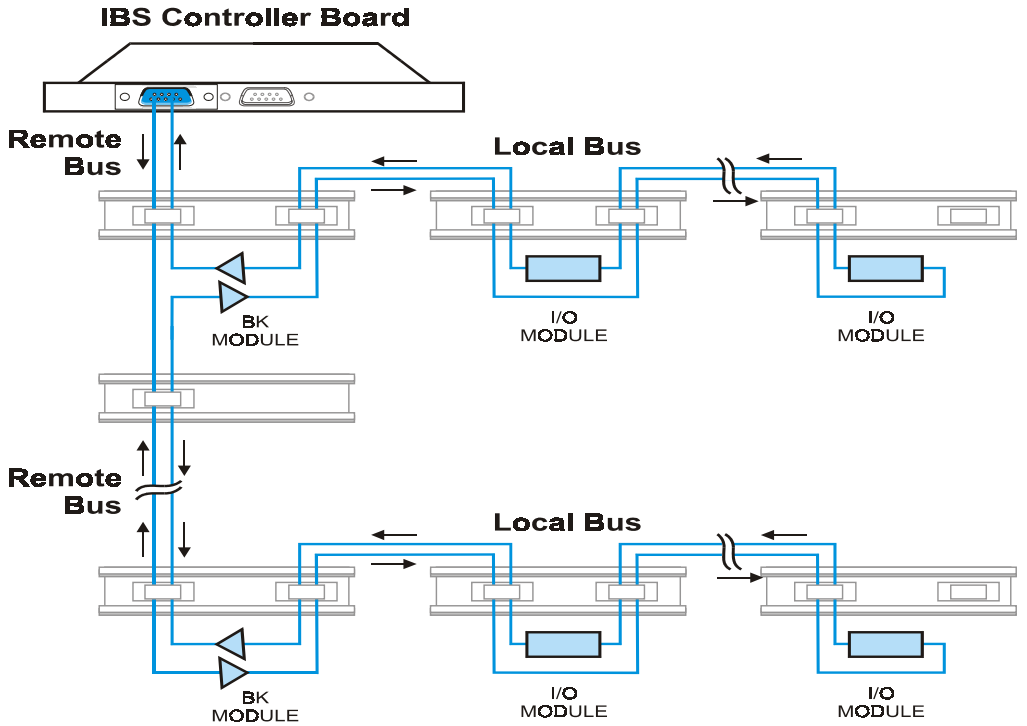


Figure 12.18
Interbus-S layout example

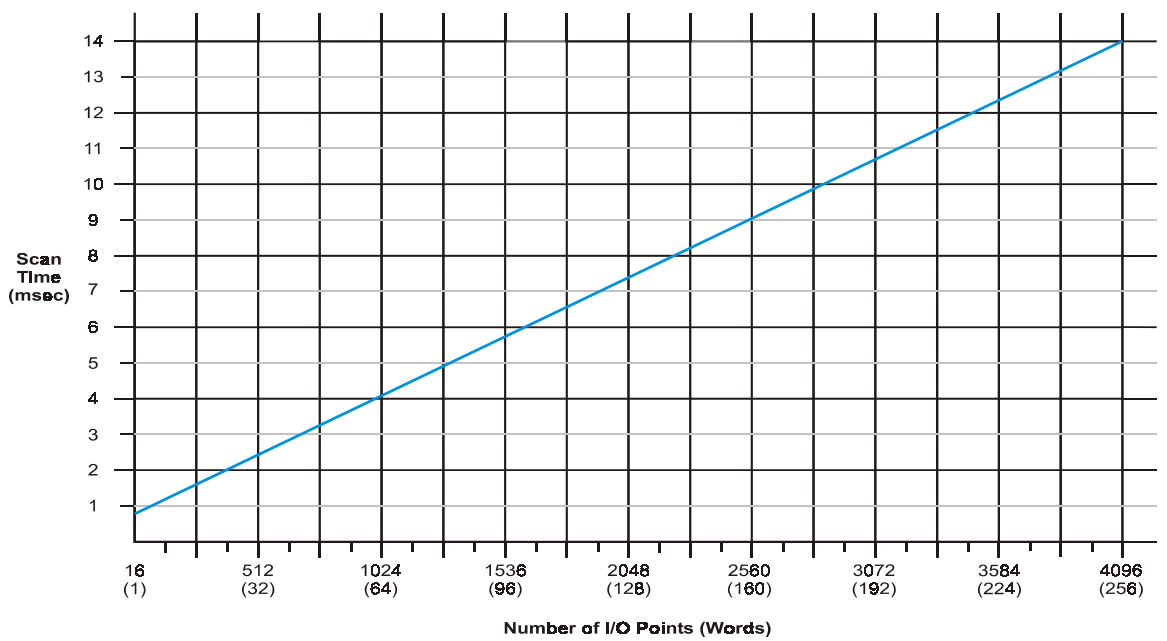


Figure 12.19
Interbus-S scan cycles and I/O points

Communication is performed through scan cycles. Each scan cycle shifts messages through each station in increasing order through the network. Data is read from the message and written to the message during each cycle making for very fast response times (see Figure 12.19).

The data link layer

The data link layer protocol provides for full duplex transmission. The complete message frame is clocked through the network on each cycle. No arbitration or contention for access to the network is required since each station has access during each cycle. All input and output data are updated and transferred during each scan cycle. CRC error checking is performed between each network connection allowing identification of the error source. Both digital and analog data are supported as well as client/server messaging.

The network does individual station addressing automatically during initialization of the network, eliminating the need to manually assign these during system startup. This is accomplished through an identification (ID) cycle that tells the controller the type and physical order of location of the stations on the network (see Figure 12.20).

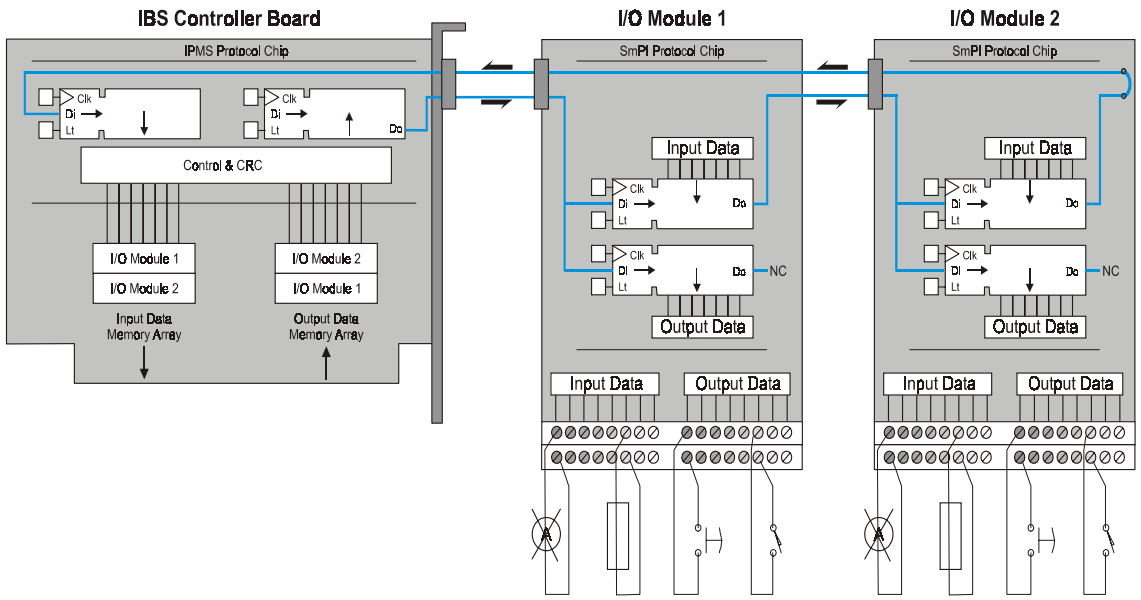


Figure 12.20
Interbus block diagram example

The application layer

Interbus-S supports network and module diagnostics and monitoring through a special ‘telegram’ message sent out after each byte is shifted. All stations monitor this message simultaneously. This unique function is accomplished by a ‘telegram control’ switch in each station (see Figure 12.21) which automatically activates after each 8-bit shift allowing not only simultaneous reception of the message by all stations, but synchronization information as well.

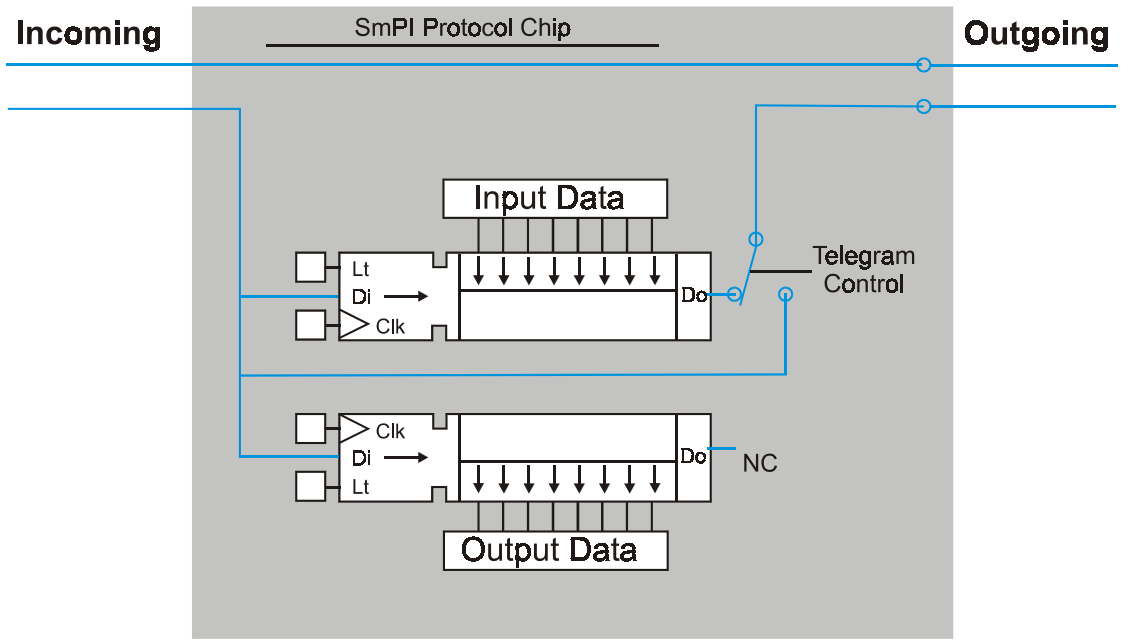


Figure 12.21
Interbus-S register flow diagram

Figure 12.22 shows the ID and scan transmission frames noting the location of the various field parameters.

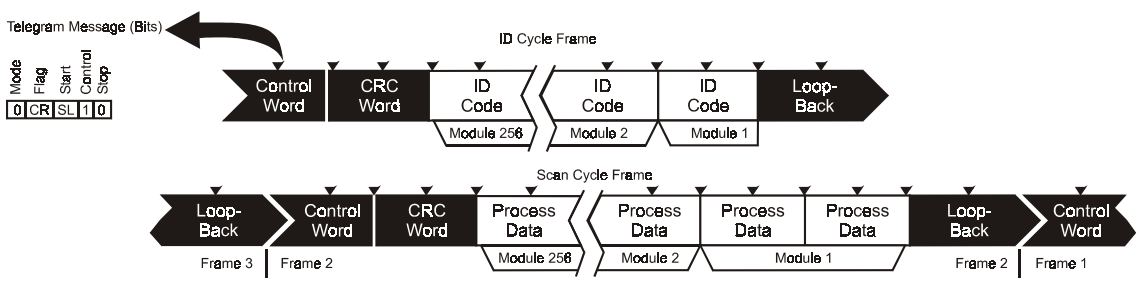


Figure 12.22
Interbus-S packet configuration

12.7 Profibus

Introduction to Profibus

Profibus is an open standard Fieldbus defined by the German DIN 19245 Parts 1 and 2. It is based on a token bus/floating master system. There are three different types of Profibus – FMS, DP and PA. Fieldbus message specification (FMS) is used for general data acquisition systems. DP is used when fast communications are needed. And PA is used in areas when intrinsically safe devices and intrinsically safe communications are needed. Figure 12.23 outlines the structure of the various versions of Profibus.

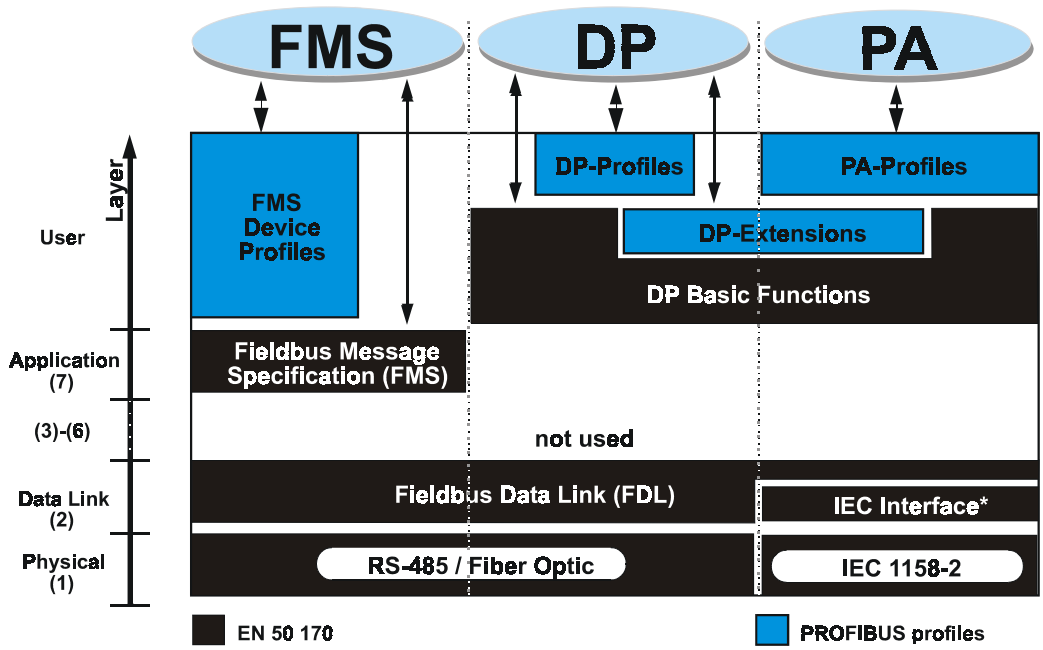


Figure 12.23
Profibus protocol architecture

The physical layer

The physical layer specifies the type of Profibus transmission medium. The RS-485 voltage standard is defined for the FMS and DP versions of Profibus. The IEC 1158-2 standard is used in the PA version. For FMS and DP a maximum number of 255 stations are possible.

• FMS (RS-485):	187.5 kbps	General use
• DP (RS-485):	500 kbps /1.5 Mbps/12 Mbps	Fast devices
• PA (IEC 1158-2):	31.25 kbps	Intrinsically safe

Basic properties of the RS-485 voltage standard for Profibus

Topology:	Linear bus, terminated at both ends
Medium:	Twisted pair shielded cable
Wire size:	18 AWG (0.8 mm)
Attenuation:	3 dB/km at 39 kHz
Number of stations:	32 stations without repeaters extendible to 127
Bus length:	max. 1200 meters (3940 feet) extendible to 4800 meters (7900 feet) at slow rates
Speed:	1200 to 12 Mbps
Connector:	Phoenix type screw or 9-pin D-sub connector

IEC 1158-2 is a standardized current standard used in special areas of a factory or plant that require intrinsically safe devices. IEC 1158-2 works by modulating a Manchester encoded bipolar NRZ ±10 mA signal on top of a 9 to 32 dc voltage. This 10 mA creates a ±1 volt signal that is read by each of the devices on the bus.

It is very easy to connect Profibus FMS, DP and PA versions together on the same system, as the main difference between the FMS, DP and PA versions is the physical layer. This allows a company to run lower cost devices in most of the plant (FMS), fast devices (DP) in parts of the plant that need the speed. Intrinsically safe devices (PA) are used in the areas of the plant that need intrinsically safe devices.

The data link layer

The data link layer is defined by Profibus as the Fieldbus data link Layer (FDL).

The medium access control (MAC) part of the FDL defines when a station may transmit data. The MAC ensures that only one station transmits data at any given time.

Profibus communication is termed hybrid medium access. It uses two methods of operation:

- Token passing
- Master/slave.

By means of a software token, the token passing method ensures (which when passed from node to node assigns the right of transmission to that node), the assignment of the bus access right within a precisely defined time interval. This is circulated with a maximum (and configurable) token rotation time between all masters. Token passing is especially useful for communication between complex automation masters who require equal rights on the bus. The token is passed in a defined sequence (in order of increasing addresses).

The master/slave method allows the master that currently has the token to communicate with the associated slave devices. The master can then read from or write data to the slave devices.

A typical configuration is shown in Figure 12.24.

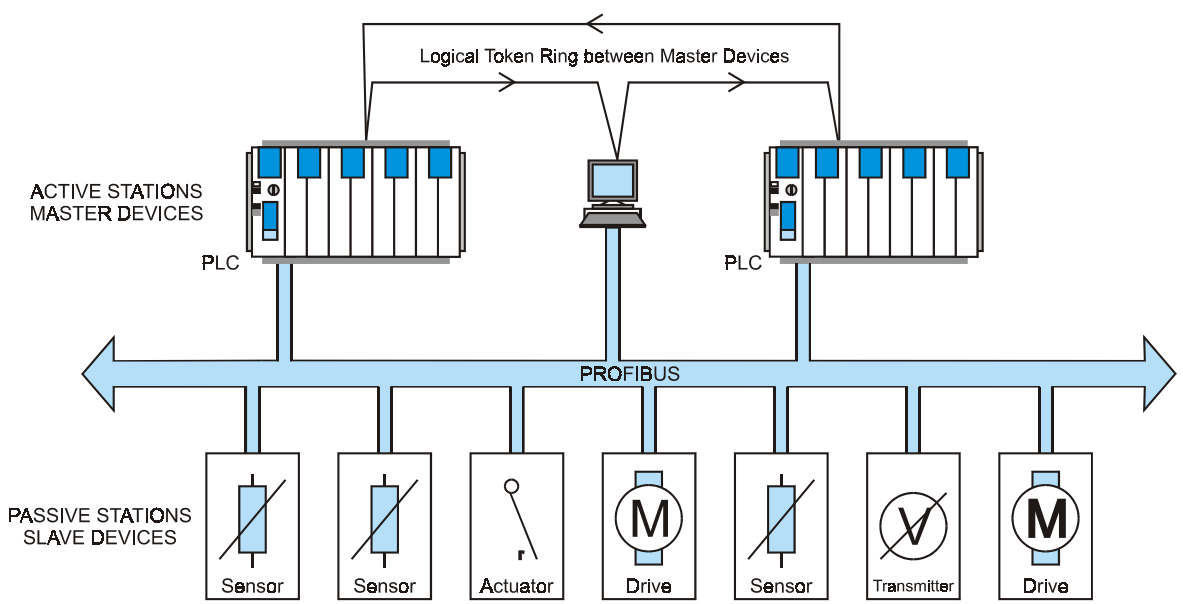


Figure 12.24
Typical architecture of a Profibus system

During the Bus system startup phase, the task of the active station's MAC is to detect logical assignment and establish the token ring. The MAC also handles adding or deleting stations (which have become inactive), deleting multiple nodes with the same address and multiple or lost tokens.

The application layer

This consists of two sections:

- Fieldbus message specification (FMS)
- Lower layer interface (LLI)

The application layer is defined in DIN 19245 part 2.

The Profibus communication model

The part of the application process in a field device that is readable for communication is called the virtual field device (VFD). The VFD contains the communication objects that may be manipulated by the services of the application layers. The objects of a real device that are readable for the communication (variables, programs, data domains) are called communication objects.

All communication objects of a Profibus station are entered into its local object dictionary (source OD). There are two types:

- Static communication objects
- Dynamic communication objects

Static communication objects are defined in the static object dictionary. They may be predefined by the manufacturer of the device, or defined during the configuration of the bus system. Static communication objects are used mainly for communication in the field area. Profibus recognizes the following static communication objects:

- Simple variable
- Array – sequence of simple variables of the same type
- Record – sequence of simple variables, not necessarily of the same type
- Domain – data range
- Event

Dynamic communication objects are entered into the dynamic part of the OD (list of variable lists of program invocations). They may be predefined or defined, deleted or changed by the application services in the operational phase.

Profibus supports the following dynamic communication objects:

- Program invocation
- Variable list (sequence of simple variables, arrays or records).

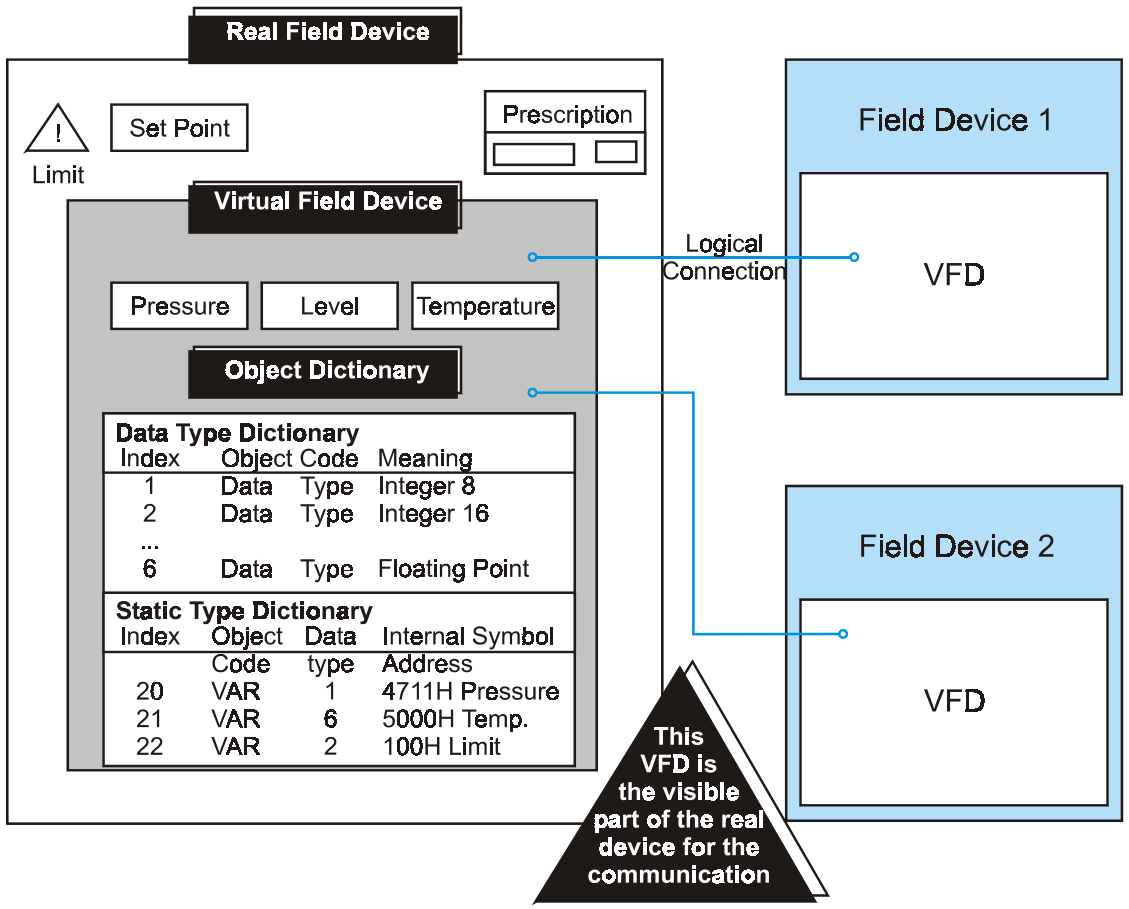


Figure 12.25
Virtual field device (VFFigD) with object dictionary (OD)

There are two methods for accessing the variables:

- Addressing by name (using a symbolic name)
- Physical addressing (to access a physical location in memory)

Profibus defines logical addressing (by symbolic name) as the preferred method as this increases the speed of access.

Application services

From the point of view of an application process, the communication system is a service provider offering various application services – the FMS services. The FMS describes the communication objects, the application services, and the resulting models from the viewpoint of the communication partner. There are two types of services:

- Confirmed services: These are only permitted on connection-oriented communication relationships.
- Unconfirmed services: These are used on connectionless communication relationships such as broadcast and multicast.

Note:
Refer to ‘Connectionless-oriented’ and ‘Connection-oriented’ under Lower layer interface following for an explanation of these terms.

Service primitives in the Profibus standard describe the execution of the services. The services can be divided into the following groups:

- Context management services allow establishment and release of logical connections
- Variable access services permit access to simple variables, records, arrays and variable lists
- Domain management services enable the transmission of contiguous memory areas
- Program invocation management services allow the control of program execution
- Event management services make the transmission of alarm messages possible
- VFD support services permit device identification and status report
- OD management services permit object dictionaries to be read and written

Lower layer interface (LLI)

The LLI conducts the data flow control and connection monitoring as well as the mapping of the FMS services onto the layer 2 with consideration of the various types of devices.

The user communicates with other application processes over the logic channels, the communication relationships. For the execution of the FMS and FMA7 services the LLI provides various types of communication relationships.

There are two types of communication relationships:

- **Connection-oriented relationships**
  This requires a connection establishment phase (or initiate service) before the connection can be used for data transmission. When the connection is no longer required, it may be released with the abort service (or connection release phase). The connection attribute distinguishes between defined connections where the communication partner is fixed at a configuration time and may not be changed and an open connection where the communication partner is dynamically defined in the connection establishment phase.

- **Connectionless-oriented relationships**
  Cyclic data transfer means exactly one variable is permanently read or written over a connection. A typical application for cyclic data transfer is the periodic update of the remove inputs and outputs of a PLC. Acyclic data transfer means an application sporadically accesses various communication objects over a connection.

Communication relationship list (CRL)

The CRL contains the description of all communication relationships of a device independent of the time of their usage.

Network management

In addition to the application services and FMS models, Profibus includes specifications for network management (Fieldbus management layer 7, FMA7).

FMA7 functions are defined in three groups:

- **Context management**
  This allows the establishment and release of management connections.
- **Configuration management**
  This allows the CRL to be loaded and read, access to variable, statistic counters and parameters of the layers 1 and 2, identification of communication components of the stations and registration of stations.
- **Fault management**
  This allows the indication of faults and events and the reset of stations.

Profibus profiles

For the various application fields it is necessary to adopt the functionality actually needed for the real world. A profile includes application specific definitions of the meanings of the communication functions, as well as the interpretation of status and error indications.

Profiles for the following application fields are available:

- Building automation
- Drive control
- Sensors and actuators
- Programmable logic controllers
- Textile machines

These enable different manufacturers, which use the same profile, to have full interoperability with the different devices on a common interconnecting Profibus.

Gateway

Gateways are required to link other protocols to the Profibus system. Some gateways are easy to implement, such as one from Profibus to the higher-level MAP. This is due to both standards adhering to the OSI model and the good relationship of Profibus to the definitions of the MAP layer 7 functions (using MMS).

Implementation features

No special hardware components are required to implement the Profibus protocol, providing the microprocessor has a UART serial interface. The implementation can range from a simple slave device, which has one microprocessor (such as an Intel 8051). This executes both the protocol and the application task; to a complex master device, which has communication, functions (and protocol) implemented on one processor. The application tasks are performed on a separate processor. For time critical applications, it is possible to implement the functions of layers 1 and 2 using a special hardware circuit (e.g. AS-iCs or a Motorola 68302).

12.8 Factory information bus (FIP)

The FIP is the result of work carried out by companies located primarily in France, Italy and Belgium. US companies such as Honeywell are involved with French manufacturers in developing the World FIP standard (see next section).

The FIP standard aims for very high transmission rates and strictly defined scanning intervals.

Bus access method

The broadcasting approach is used, with a central unit (called the bus arbitrator) co-coordinating the transmissions. This means that it is not necessary to give each device a unique address. A variable (processed by the transmitter only) is transmitted on the bus by one transmitter and is read by any number of receivers situated on the same bus.

The bus arbitrator has three operating cycles:

- **Cyclic traffic**
  The bus arbitrator names a set of variables using a table command.
- **Aperiodic traffic**
  The bus arbitrator calls on request variable from every device.
- **Message service**
  The arbitrator gives the right to transmit to a device, which requested it during the previous cyclic traffic period.

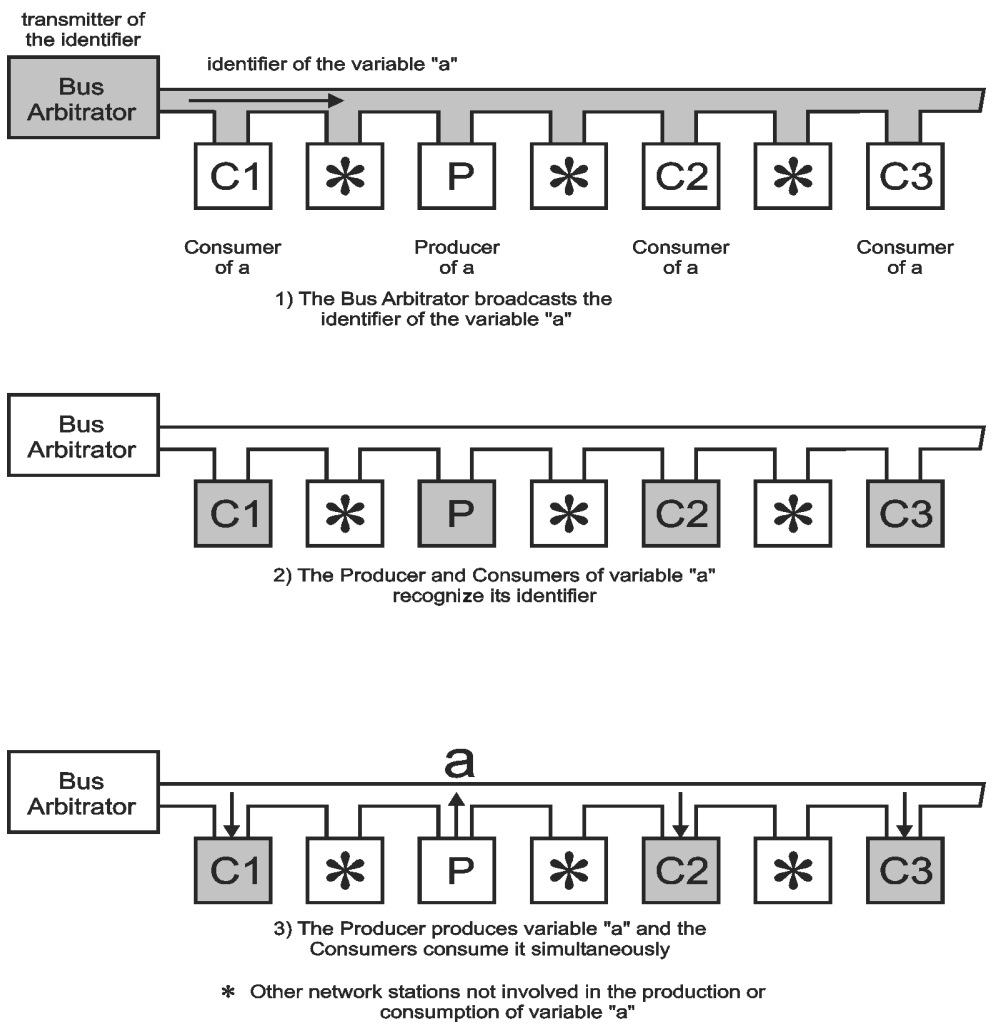


Figure 12.26
Operation of FIP

The physical layer

The FIP standard allows twisted pairs, optical fiber or coaxial cable in a bus topology to a maximum distance of 2 km. Speeds can vary between 31.25 and 1000 kbps. A maximum number of 256 devices is allowed on the bus network.

The data link layer

The data link layer is non-proprietary FIP.

Installations

There are a number of installations in France and Italy that are being used to evaluate the FIP standard.

The FIP standard has evolved into the WorldFIP standard as discussed in the next section.

12.9 WorldFip

WorldFip is the present day association of European manufacturers that support the use and international standardization of the factory information protocol (FIP). FIP is a communications protocol developed and presently in use in Europe.

The FIP physical layer is compliant with the IEC S50.02, which allows for twisted pair, or fiber optic cable media operating at 31.25 kbps, 1, or 2.5 Mbps, designated respectively as S1, S2 and S3. S2 is the standard speed. An additional speed of 5 Mbps has been designated for fiber optic media. Devices can be bus powered or independently powered.

FIP uses a producer-distributor-consumer type communications and access control model for transferring time critical information throughout the network. Devices and their variables are designated either producers or consumers of specific variables. One device can be both a producer of one variable but a consumer of another variable located somewhere else on the network.

Instead of a poll and response type integration of the entire network and then routing the required information to the specified destination, the FIP bus arbitrator simply places the request for a variable on to the network in a broadcast fashion. All devices ‘hear’ the broadcast. The producer of that variable then places it on the network again in a broadcast form. It is then available to all consumers of that particular variable – see Figure 12.26. This procedure allows rapid access of all variables in a timely and determined manner while ensuring no collisions and therefore a very efficient use of the network capabilities.

This requires a configuration and scheduling table within the arbitrator and the devices. Certain variables may need to be polled more often than others and this is taken into account in the scheduling table. In fact, the table can be configured for the specific application and time requirements of the process, making FIP very adaptable to changing conditions and new applications. The table is defined during initial network configuration. An example table is shown in Figure 12.27 for reference.

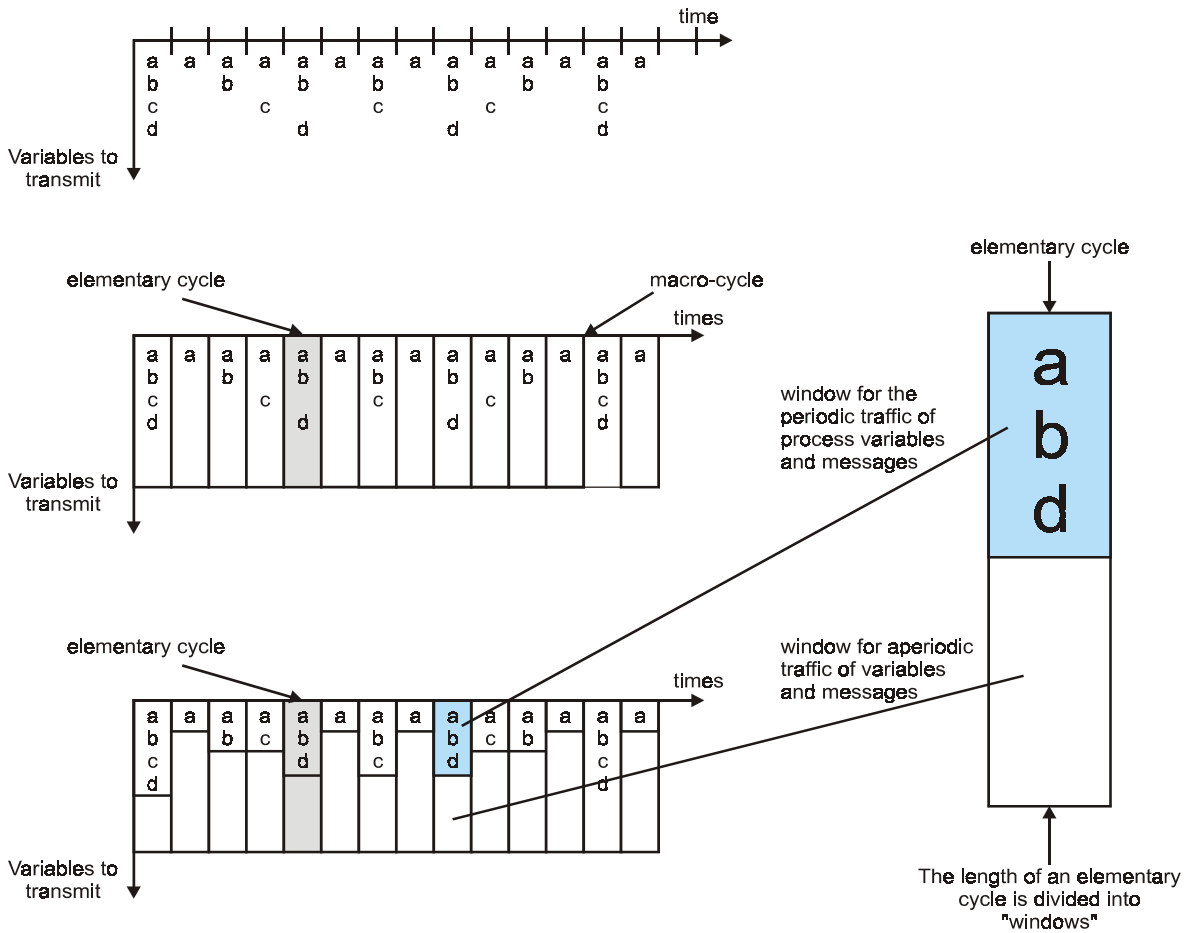


Figure 12.27 FIP
Configuration and scheduling table

FIP uses the Manchester coding to transfer data and synchronizing information. The unique frame start and stop sequences are used to help the receivers distinguish clearly the start and end of the data frames from random noise that may occur on the network. The unusual pattern is clearly different than almost any randomly occurring noise pattern.

12.10 Foundation Fieldbus

Introduction to Foundation Fieldbus

Prior to 1994, two organizations were independently trying to develop a communication solution standard for the process control industry. These organizations were WorldFip (North America) and Interoperable Systems Project (ISP). In September 1994 these two organizations merged to become the Foundation Fieldbus (FF).

Foundation Fieldbus takes full advantage of the emerging ‘smart’ field devices and modern digital communications technology allowing end user benefits such as:

- Reduced wiring
- Communications of multiple process variables from a single instrument
- Advanced diagnostics
- Interoperability between devices of different manufacturers

- Enhanced field level control
- Reduced startup time
- Simpler integration

The concept behind Foundation Fieldbus is to preserve the desirable features of the present 4–20 mA standard (such as a standardized interface to the communications link, bus power derived from the link and intrinsic safety options) while taking advantage of the new digital technologies. This will provide the features noted above due to the:

- Reduced wiring due to the multidrop capability
- Flexibility of supplier choices due to interoperability
- Reduced control room equipment due to distribution of control functions to the device level
- Increased data integrity and reliability due to the application of digital communications.

To understand how this standard works, it is helpful to look at Foundation Fieldbus in terms of the OSI Model. The FF consists of three parts that correspond to OSI layers 1, 2, 7 and 8. Layer 8 of the OSI model corresponds to the 'user' layer.

The physical layer and wiring rules

The physical layer standard has been approved and is detailed in the IEC 1158-2 and the ISA standard S50.02-1992. It supports communication rates of 31.25 kbps and 10 Mbps. All of these use the Manchester bi-phase L encoding scheme with four encoding states as shown in Figure 12.28. The use of the N+ and N– encoding states is illustrated in Figure 12.29. Devices can be optionally powered from the bus under certain conditions as detailed below for the various configurations. The 31.25 kbps (or H1, or low-speed bus) can support from 2 to 32 devices that are not bus powered, two to twelve devices that are bus powered or two to six devices that are bus powered in an intrinsically safe area. Repeaters are allowed and will increase the length and number of devices that can be put on the bus. The H2 or high speed bus options are not currently being implemented, but have been superseded by the high speed Ethernet (HSE) standard. This is discussed later in this section.

The low speed bus was intended to utilize existing plant wiring and is referred to as Type B wiring (shielded twisted pair) and with #22 AWG can be used for segments up to 1200 m (3936 feet). The higher speeds require higher grade cabling and are referred to as Type A. For Type A cable (shielded twisted pair) for H1 #18 AWG can be used up to 1900 meters (6232 feet). Two additional types of cabling are specified and are referred to as Type C (multi-pair twisted without shield) and Type D (called multi-core, no shield). Type C using #26 AWG cable is limited to 400 meters (1312 feet) per segment and Type D with #16 AWG is restricted to segments less than 200 meters (660 feet).

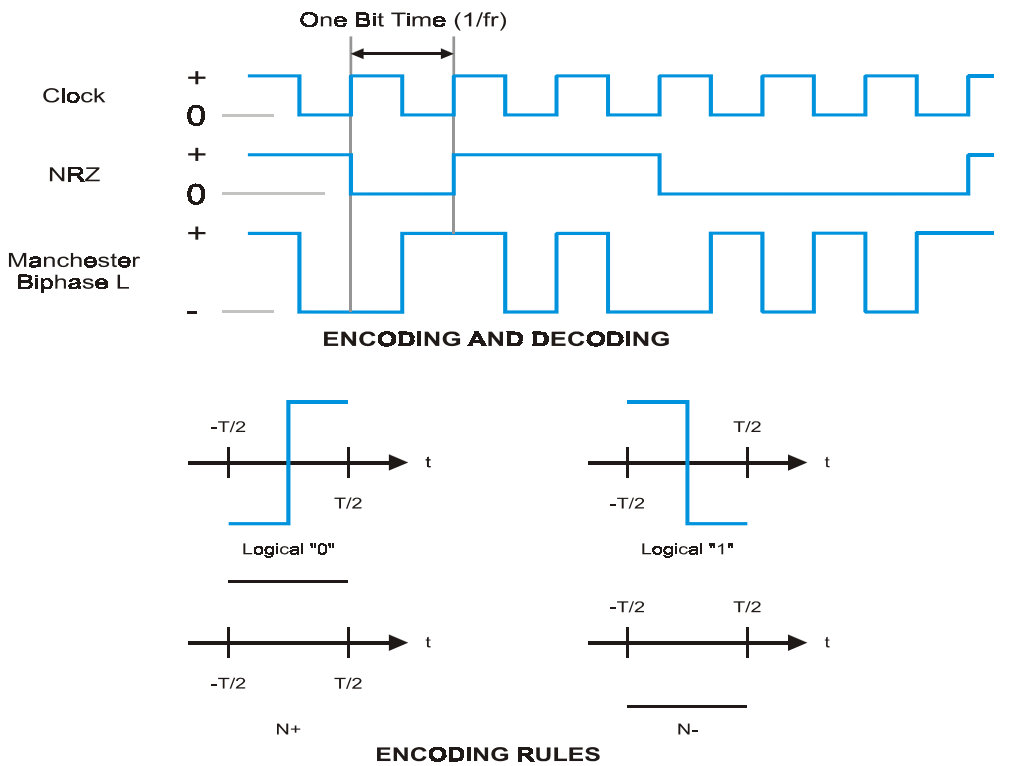
- Type A #18 AWG 1900 m (6232 feet)
- Type B #22 AWG 1200 m (3936 feet)
- Type C #26 AWG 400 m (1312 feet)
- Type D #16 AWG multi-core 200 m (660 feet)

The Foundation Fieldbus wiring method is floating balanced and equipped with a termination resistor combination connected across each end of the transmission line. Neither of the wires should ever be connected to ground. The terminator consists of a

100 Ω quarter watt resistor and a capacitor sized to pass 31.25 kHz. As an option one of the terminators can be center tapped and grounded to prevent voltage build-up on the Fieldbus. Power supplies must be impedance matched for FF. Off the shelf power supplies must be conditioned. If a 'normal power supply' is placed across the line it will load down the line due to its low impedance. This will cause the transmitters to stop transmitting.

Fast response times for the bus are one of the FF goals. For example, at 31.25 kbps on the H1 bus response times as low as 32 microseconds are expected (this will vary based on the loading of the system but will average between 32 ms and 2.2 ms with an average approximately 1 ms).

Spurs can be connected to the 'home run'. The length of the spurs depends on the type of wire used and the number of spurs connected. The maximum length is the total length of the spurs and the home run.



Symbols		Encoding
1	(ONE)	Hi-Lo transition (mid-bit)
0	(ZERO)	Lo-Hi transition (mid-bit)
N+	(NON-DATA PLUS)	Hi (No transition)
N-	(NON-DATA MINUS)	Lo (No transition)

ENCODING RULES

Figure 12.28
Foundation Fieldbus physical layer

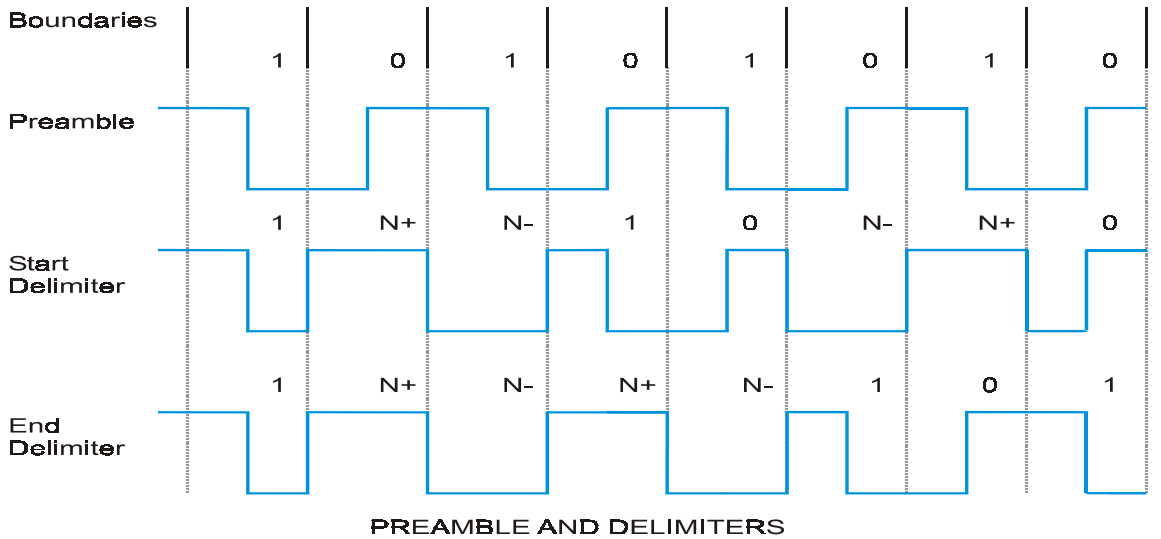


Figure 12.29
Use of N+ and N– encoding states

The physical layer standard has been out for some time. Most of the recent work has been focused on these upper layers and are defined by the FF as the 'communications stack' and the 'user layer'. The following sections will explore these upper layers. Figure 12.30 helps understanding the subsequent discussions.

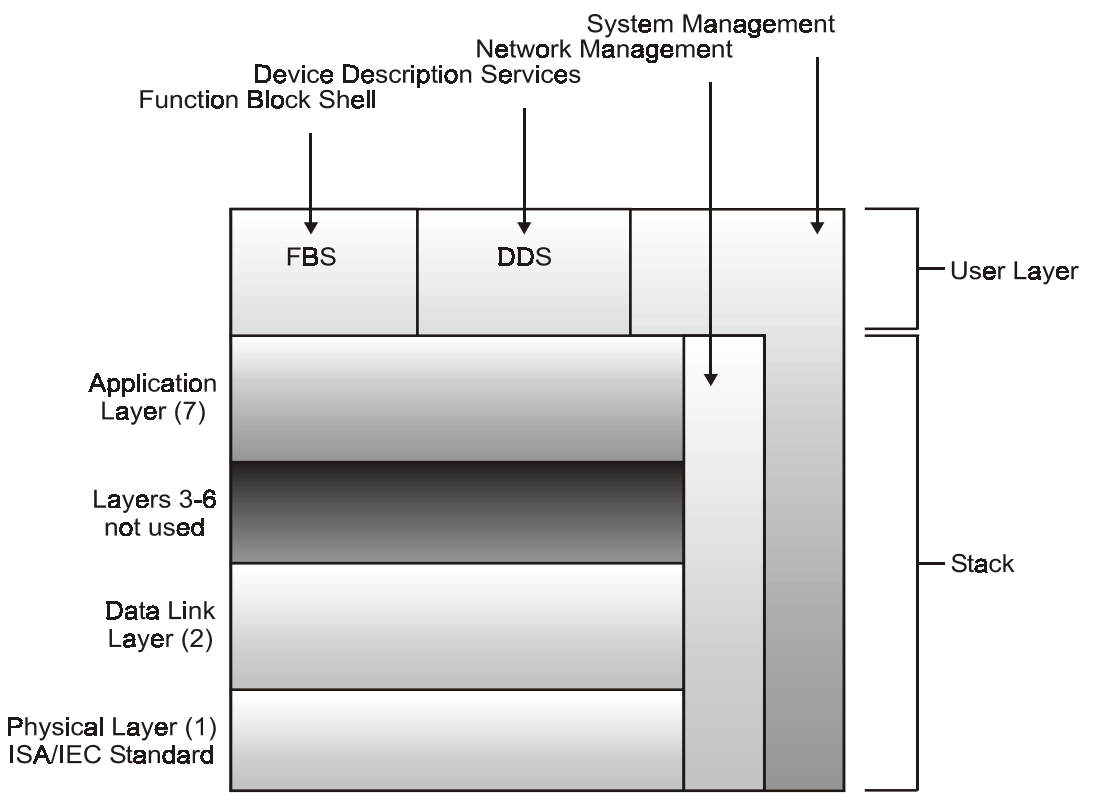


Figure 12.30
The OSI model of the FF protocol stack

The data link layer

The communications stack as defined by the FF corresponds to OSI layers two and seven, the data link and applications layers. The DLL (data link layer) controls access to the bus through a centralized bus scheduler called the link active scheduler (LAS). The DLL packet format is shown in Figure 12.31.

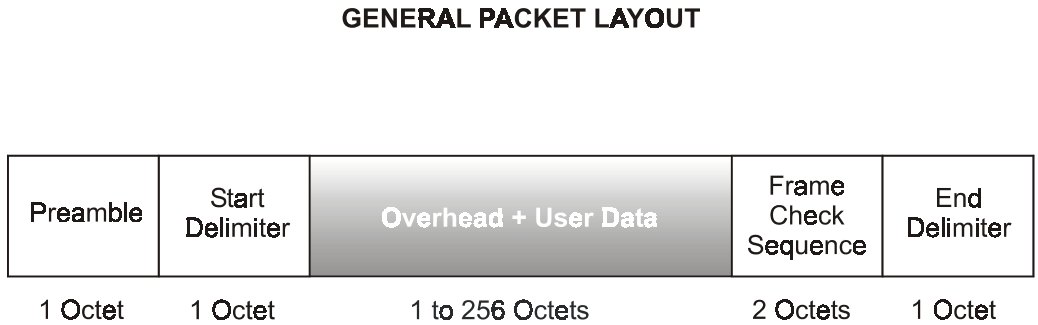


Figure 12.31
Data link layer packet format

The link active scheduler (LAS) controls access to the bus by granting permission to each device according to predefined 'schedules'. No device may access the bus without LAS permission. There are two types of schedules implemented: cyclic (scheduled) and acyclic (unscheduled). It may seem odd that one could have an unscheduled 'schedule', but these terms actually refer to messages that have a periodic or non-periodic routine, or 'schedule'.

The cyclic messages are used for information (process and control variables) that require regular, periodic updating between devices on the bus. The technique used for information transfer on the bus is known as the publisher–subscriber method. Based on the user predefined (programmed) schedule the LAS grant permission for each device in turn access to the bus. Once the device receives permission to access the bus it 'publishes' its available information. All other devices can then listen to the 'published' information and read it into memory (subscribe) if it requires it for its own use. Devices not requiring specific data simply ignore the 'published' information.

The acyclic messages are used for special cases that may not occur on a regular basis. These may be alarm acknowledgment or special commands such as retrieving diagnostic information from a specific device on the bus. The LAS detects time slots available between cyclic messages and uses these to send the acyclic messages.

The application layer

The application layer in the FF specification is divided into two sub-layers – the Foundation Fieldbus access sublayer (FAS) and the Foundation Fieldbus messaging specification (FMS).

The capability to pre-program the 'schedule' in the LAS provides a powerful configuration tool for the end user since the time of rotation between devices can be established and critical devices can be 'scheduled' more frequently to provide a form of prioritization of specific I/O points. This is the responsibility and capability of the FAS. Programming the schedule via the FAS allows the option of implementing (actually, simulating) various 'services' between the LAS and the devices on the bus.

Three such 'services' are readily apparent such as:

- Client/server: with a dedicated client (the LAS) and several servers (the bus devices)
- Publisher/subscriber: as described above, *and*
- Event distribution: with devices reporting only in response to a 'trigger' event, or by exception, or other predefined criteria.

These variations, of course, depend on the actual application and one scheme would not necessarily be 'right' for all applications, but the flexibility of the Foundation Fieldbus is easily understood from this example.

The second sub-layer, the Foundation Fieldbus messaging specification (FMS), contains an 'object dictionary' which is a type of database that allows access to Foundation Fieldbus data by tag name or an index number. The object dictionary contains complete listings of all data types, data type descriptions, and communication objects used by the application. The services allow the object dictionary (application database) to be accessed and manipulated. Information can be read from or written to the object dictionary allowing manipulation of the application and the services provided.

The user layer

The FF specifies an eighth layer called the user layer that resides 'above' the application layer of the OSI model; this layer is usually referred to as Layer 8. In the Foundation Fieldbus this layer is responsible for three main tasks – network management, system management and function block/device description services. Figure 12.32 illustrates how all the layer's information packets are passed to the physical layer.

The network management service provides access to the other layers for performance monitoring and managing communications between the layers and between remote objects (objects on the bus). The system management takes care of device address assignment, application clock synchronization, and function block scheduling. This is essentially the time coordination between devices and the software, and ensures correct time stamping of events throughout the bus.

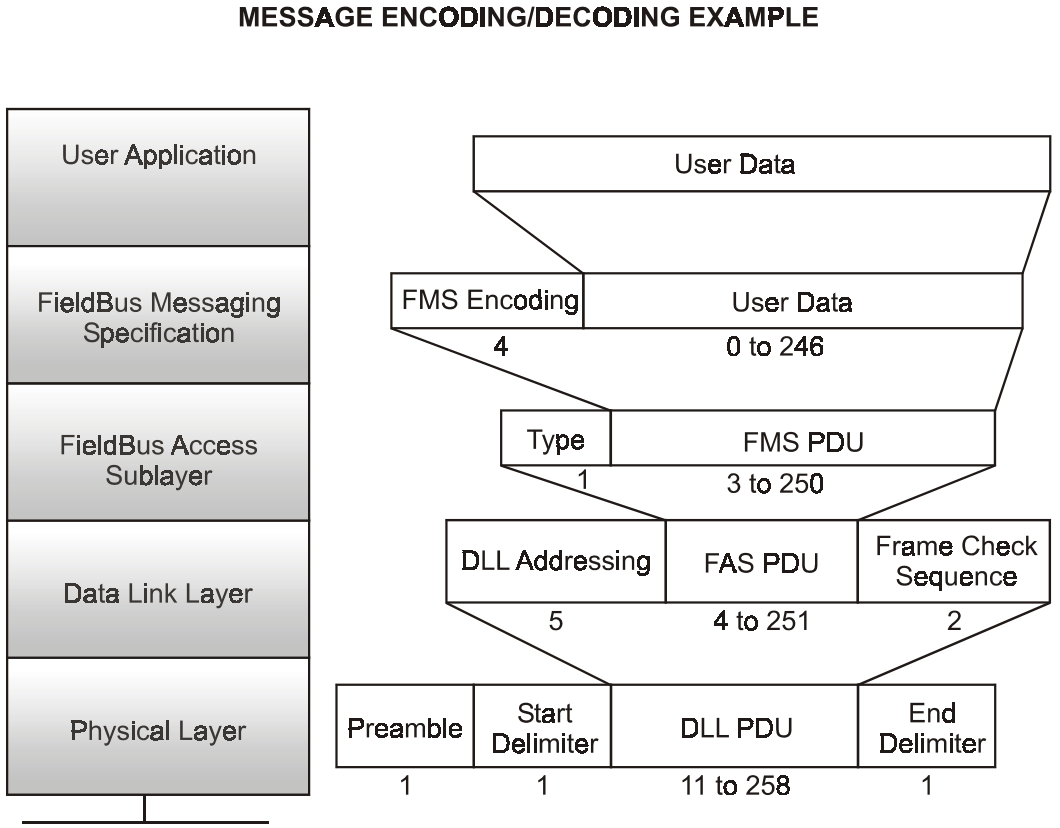


Figure 12.32
The passage of information packets to the physical layer

Function blocks and device description services provide pre-programmed ‘blocks’, which can be used by the end user to eliminate redundant and time-consuming configuration. The block concept allows selection of generic functions, algorithms, and even generic devices from a library of objects during system configuration and programming. This process can dramatically reduce configuration time since large ‘blocks’ are already configured and simply need to be selected. The goal is to provide an open system that supports interoperability and a device description language (DDL), which will enable multiple vendors, and devices to be described as ‘blocks’ or ‘symbols’. The user would select generic devices then refine this selection by selecting a DDL object to specify a specific vendor’s product. Entering a control loop ‘block’ with the appropriate parameters would nearly complete the initial configuration for the loop. Advanced control functions and mathematics ‘blocks’ are also available for more advanced control applications.

Error detection and diagnostics

FF has been developed as a purely digital communications bus for the process industry and incorporates error detection and diagnostic information. It uses multiple vendors’ components and has extensive diagnostics across the stack from the physical link up through the network and system management layers by design.

The signal method used by the physical layer timing and synchronization is monitored constantly as part of the communications. Repeated messages and the reason for the repeat can be logged and displayed for interpretation.

In the upper layer, network and system management is an integral feature of the diagnostic routines. This allows the system manager to analyze the network 'on-line' and maintain traffic loading information. As devices are added and removed, optimization of the link active scheduler (LAS) routine allows communications optimization dynamically without requiring a complete network shutdown. This ensures optimal timing and device reporting, giving more time to higher priority devices and removing, or minimizing, redundant or low priority messaging.

With the device description (DD) library for each device stored in the host controller (a requirement for true interoperability between vendors) all the diagnostic capability of each vendors' produces can be accurately reported and logged and/or alarmed to provide continuous monitoring of each device.

High-speed Ethernet (HSE)

High-speed Ethernet (HSE) is the Fieldbus Foundation's backbone network running at 100 Mbps. HSE Field Devices are connected to the backbone via HSE linking devices. A HSE linking device is a device used to interconnect H1 Fieldbus segments to HSE to create a larger network. A HSE switch is an Ethernet device used to interconnect multiple HSE devices such as HSE linking devices and HSE field devices to form an even larger HSE network. HSE hosts are used to configure and monitor the linking devices and H1 devices. Each H1 segment has its own link active scheduler (LAS) located in a linking device. This feature enables the H1 segments to continue operating even if the hosts are disconnected from the HSE backbone. Multiple H1 (31.25 kbps) Fieldbus segments can be connected to the HSE backbone via linking devices.

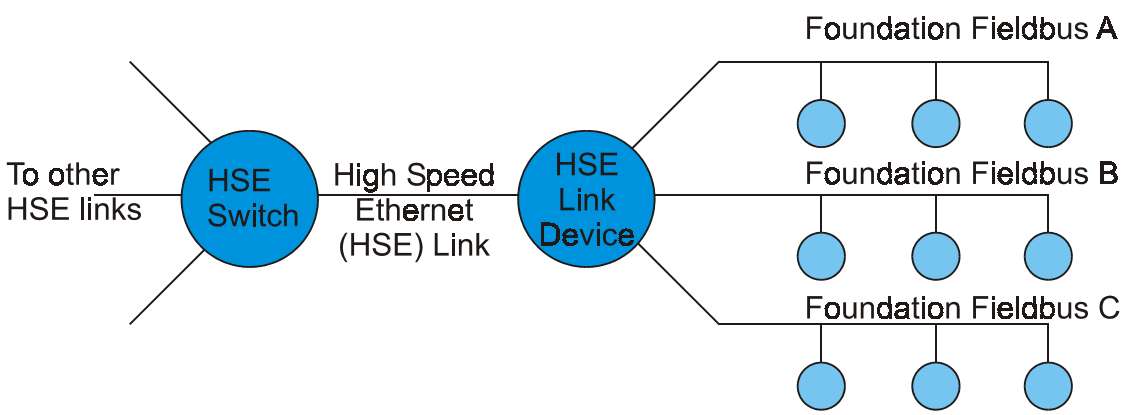


Figure 12.33
High-speed Ethernet and Foundation Fieldbus